Design and Simulation of a 2026 FIA Formula 1 Hybrid Powertrain Vehicle: Architecture, Electrical Systems, and Lap Performance Analysis

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Abstract

The 2026 FIA Technical Regulations introduce the most transformative powertrain overhaul in the hybrid era of Formula 1, mandating a 350 kW Motor Generator Unit–Kinetic (MGU-K)—nearly triple the outgoing 120 kW specification—while simultaneously eliminating the Motor Generator Unit–Heat (MGU-H) and requiring the exclusive use of fully sustainable fuel. These regulatory changes fundamentally rebalance the power-unit architecture, elevating the electrical drivetrain from a secondary energy-recovery device to a primary propulsive element that contributes approximately 43% of total system power. This paper presents a complete ground-up concept design of a 2026-regulation-compliant Formula 1 car encompassing vehicle architecture—3580 mm wheelbase, 800 kg minimum mass, and a 45 F / 55 R weight distribution—alongside a detailed electrical-system design featuring a 180S2P nickel–manganese–cobalt (NMC) battery pack operating at 648 V nominal, a field-oriented control (FOC) strategy with space-vector pulse-width modulation (SVPWM) for the inverter stage, and an extended Kalman filter for real-time state-of-charge (SOC) estimation. The removal of the MGU-H cooling load motivates a minimalist undercut sidepod aero-thermal philosophy that reduces frontal area and drag. A point-mass lap time simulator implemented in MATLAB models energy deployment around the Autodromo Nazionale Monza, subject to SOC constraints of 20–95%, a hard energy-budget gate of 3 MJ, and an end-to-end drivetrain efficiency chain of 90.3%. Simulation results predict a lap time of 1:16.79—a 2.3 s improvement over the 2019 Monza lap record—with a peak speed of 348.2 $\mathrm{km\,h^{-1}}$, total energy deployed of 4.005 MJ, and a minimum SOC of 25.3%. These findings underscore the decisive shift of the engineering battleground toward the inverter, battery-management, and motor-control subsystems. Limitations of the present study and avenues for future work—including multi-lap race simulation, a full 14-degree-of-freedom vehicle dynamics model, and Reynolds-averaged Navier–Stokes (RANS) computational fluid dynamics validation—are discussed.

Keywords: Formula 1, hybrid powertrain, MGU-K, battery management system, lap time simulation, field-oriented control, 2026 FIA regulations

Contents

1 Introduction

The evolution of propulsion systems in Formula 1 represents a continuous pursuit of peak thermodynamic efficiency intertwined with the demands of highly competitive motorsport. Historically, the sport relied entirely on high-revving, naturally aspirated internal combustion engines (ICE) that prioritized peak power output over fuel efficiency. However, a sweeping paradigm shift occurred in 2014 with the introduction of the V6 turbo-hybrid era. This regulation change mandated the integration of complex Energy Recovery Systems (ERS), comprising the Motor Generator Unit-Kinetic (MGU-K) and the Motor Generator Unit-Heat (MGU-H), operating in tandem with the internal combustion engine. While this marked a monumental step toward road-relevant hybridization and record-breaking thermal efficiency, the combustion engine remained the dominant source of propulsive power, with the electrical components serving primarily as supplementary performance enhancers.

The 2026 regulatory cycle introduces a fundamental turning point in the sport's technical trajectory, decisively redefining the balance of power generation. These new regulations mandate a drastic shift toward electrical propulsion, establishing a near-equal power split between the internal combustion engine and the electrical system. The elimination of the complex MGU-H is counterbalanced by a substantial upgrade to the MGU-K, which will see its deployment capacity nearly triple to 350 kilowatts (kW). Concurrently, the energy storage system is scaled to manage an operational capacity of approximately 4 megajoules (MJ) per lap. This radical architectural restructuring dictates that the electrical powertrain transition from a supporting role to a primary driver of performance, while the ICE transitions to utilize fully sustainable drop-in fuels to further the championship's net-zero carbon objectives.

This unprecedented reliance on electrical energy introduces a myriad of profound engineering challenges. The packaging of the heavily uprated MGU-K, alongside the significantly larger and denser battery pack within the chassis, requires meticulous spatial optimization. This electrical mass significantly influences the vehicle's weight distribution, moment of inertia, and center of gravity. Furthermore, thermal management becomes an acute constraint; rapidly dissipating the immense heat flux generated by a 350 kW electrical motor and a high-cycling 4 MJ battery mandates aggressive cooling architectures that inherently conflict with aerodynamic drag reduction goals. Energy management strategies must also be entirely overhauled. The increased electrical deployment ratio necessitates highly sophisticated, algorithmic deployment and harvesting profiles to prevent severe power derating—and subsequent velocity drop-offs—at the end of long straights. Aerodynamically, the 2026 regulations mandate the implementation of active aerodynamic systems to mitigate drag and compensate for the altered power delivery characteristics, adding critical complexity to the vehicle's dynamic behavior.

The purpose of this paper is to present a comprehensive, multi-disciplinary design study of a 2026-compliant Formula 1 car. By analyzing the symbiotic relationship between the drastically altered power unit architecture and the new generative aerodynamic regulations, this study systematically explores optimal solutions for packaging, thermal management, and energy deployment. Utilizing advanced computational fluid dynamics (CFD) and transient lap time simulation (LTS), we evaluate the cascading effects of the 350 kW electrical architecture on chassis dynamics and overall vehicle concept. Ultimately, this research aims to establish a theoretical engineering framework for maximizing competitive performance under the constraints of the most electrically dominant regulatory cycle in the history of the sport.

2 2026 FIA Technical Regulations Overview

The 2026 Formula 1 technical regulations represent one of the most comprehensive and radical overhauls in the championship's history, simultaneously addressing power unit architecture, chassis design, and aerodynamic philosophy. Dictated by the Federation Internationale de l'Automobile (FIA) [1], these mandates aim to drastically reduce the carbon footprint of the sport, increase the road relevance of the technology, and enhance the competitive spectacle by enabling cars to follow each other more closely. This section delineates the core regulatory changes that govern the vehicle design space for the 2026 technical cycle.

2.1 Power Unit Regulations

The foundation of the 2026 regulations rests upon a radically revised Power Unit (PU) architecture that fundamentally alters the energy economics of the vehicle. While the internal combustion engine (ICE) remains a 1.6-liter, 90-degree V6 with a single turbocharger, its functional role has been significantly redefined [2]. Most notably, the regulations mandate the complete removal of the Motor Generator Unit-Heat (MGU-H), eliminating a highly complex, expensive, and non-road-relevant component that previously provided near-continuous electrical harvesting from exhaust gases.

To compensate for the loss of the MGU-H and aggressively expand the role of electrical propulsion, the regulations dramatically uprate the remaining Motor Generator Unit-Kinetic (MGU-K). The maximum deployment power of the MGU-K is nearly tripled to 350 kilowatts (kW), establishing a near-perfect 50/50 power distribution between the internal combustion and electrical systems. To sustain this immense electrical output, the Energy Store (ES), or battery pack, has been conceptually scaled to manage approximately 4 megajoules (MJ) of deliverable energy per lap. This necessitates exceptionally high charge and discharge rates, presenting a formidable challenge in cell chemistry, thermal management, and high-voltage control systems.

Furthermore, the 2026 regulations enforce a strict environmental mandate concerning the chemical energy source of the ICE. The championship requires the exclusive use of 100% fully sustainable, advanced "drop-in" synthetic fuels [2]. This requirement fundamentally alters combustion characteristics, fuel mass flow limitations, and thermodynamic efficiency targets, compelling powertrain engineers to optimize the ICE specifically for synthetic hydrocarbon structures while maintaining the rigorous durability requirements of the sporting regulations.

2.2 Aerodynamic Changes

Complementing the power unit overhaul is a completely reimagined aerodynamic framework designed to mitigate the inherent challenges of the new powertrain while improving the racing spectacle. The primary objective of the 2026 aerodynamic regulations is to significantly reduce overall vehicle drag to compensate for the altered power delivery profile—specifically, the potential for early electrical energy depletion and subsequent velocity drop-offs on long straights.

To achieve this, the FIA has mandated a simplification of the external bodywork [3]. The structural dimensions of the vehicle are contracted, featuring a shorter wheelbase and a narrower overall width. These dimensional reductions inherently decrease the frontal area of the vehicle.

More critically, the regulations introduce fully integrated active aerodynamics as a cornerstone of the chassis design. The vehicles will employ movable front and rear wing elements that operate in distinct configurations: a high-downforce state for cornering and a low-drag state for straight-line velocity [1]. This active system is essential for achieving the aggressive regulatory targets of reduced downforce and significantly lowered drag compared to the previous vehicle generation. Despite these reductions, the fundamental aerodynamic philosophy continues to emphasize the underbody and ground effect phenomena for downforce generation, albeit with tighter dimensional constraints on the floor and diffuser to control the wake structure and maximize the aerodynamic efficiency of the vehicle trailing behind.

2.3 Weight and Packaging Constraints

The interplay between the expanded electrical powertrain and the contracted chassis dimensions creates an exceptionally constrained packaging environment. The 2026 regulations introduce stringent mass and volumetric limitations to ensure safety, control costs, and dictate conceptual direction.

Despite the significant increase in electrical component mass—stemming from the uprated MGU-K, heavier phase cabling, and the chemically dense 4 MJ battery pack—the FIA has established a minimum weight target of 800 kg [3]. Achieving this mass limit represents one of the most profound engineering hurdles of the regulatory cycle, demanding rigorous weight optimization across all structural and mechanical assemblies to prevent performance-inhibiting mass penalties.

A critical factor contributing to overall vehicle weight management is the substantial reduction in the permissible race fuel load. Facilitated by the 50/50 power split and the inherently higher efficiency of the long electrical deployment phases, the maximum fuel mass allowed for a standard race distance is reduced precipitously from 110 kg to just 70 kg [2]. This reduction allows for a significantly smaller fuel cell, structurally altering the monocoque topology and center of gravity behind the driver.

The packaging constraints are further exacerbated by uncompromising safety mandates. The massive 4 MJ Energy Store, along with the high-voltage control electronics, must be compactly packaged within a strictly defined safety volume inside the survival cell, primarily located beneath and behind the driver [1]. The spatial integration of these components, alongside the necessity for robust, high-capacity cooling networks to manage the intense thermal rejection of the 350 kW electrical architecture, defines the core packaging challenge of the 2026 Formula 1 car. This requires an unprecedented level of spatial synergy between powertrain integration and structural chassis engineering.

3 Vehicle Architecture Overview

The fundamental architecture of this 2026-compliant Formula 1 concept is dictated by the delicate interplay between the radically revised, highly electrified power unit and the restrictive aerodynamic and packaging envelopes mandated by the FIA. The vehicle employs a traditional rear-mid engine layout, optimized specifically to accommodate the structurally significant 4 megajoule (MJ) Energy Store while adhering to the stringent 800 kg minimum weight and aggressively shortened dimensional targets. This section details the macroscopic packaging strategy, mass distribution, and primary structural subsystems that define the vehicle's dynamic baseline.

3.1 Dimensional Constraints and Mass Properties

Responding to the 2026 regulations aimed at enhancing agility and reducing overall vehicle footprint, the wheelbase is constrained exactly to 3580 mm, dictating a highly compact internal arrangement. The overall vehicle length spans 5125 mm, bounded by the strict aerodynamic dimensional boxes (Figure 1). Achieving the 800 kg minimum weight target represents a profound engineering challenge given the massive inclusion of the uprated 350 kW Motor Generator Unit-Kinetic (MGU-K) and the dense 4 MJ battery.

Table 1: Vehicle Architecture Design Parameters

Parameter	**Value**
Wheelbase	3580 mm
Overall Length	5125 mm
Vehicle Width	2000 mm
Front Track	1600 mm
Rear Track	1620 mm
CoG Height	270 mm
Weight Distribution	45% F / 55% R
Ground Clearance (F/R)	32 mm / 30 mm
Driver Recline	42°
Gearbox Length	590 mm
Total Mass	800 kg
Fuel Load	70 kg
Battery Energy	∼4 MJ

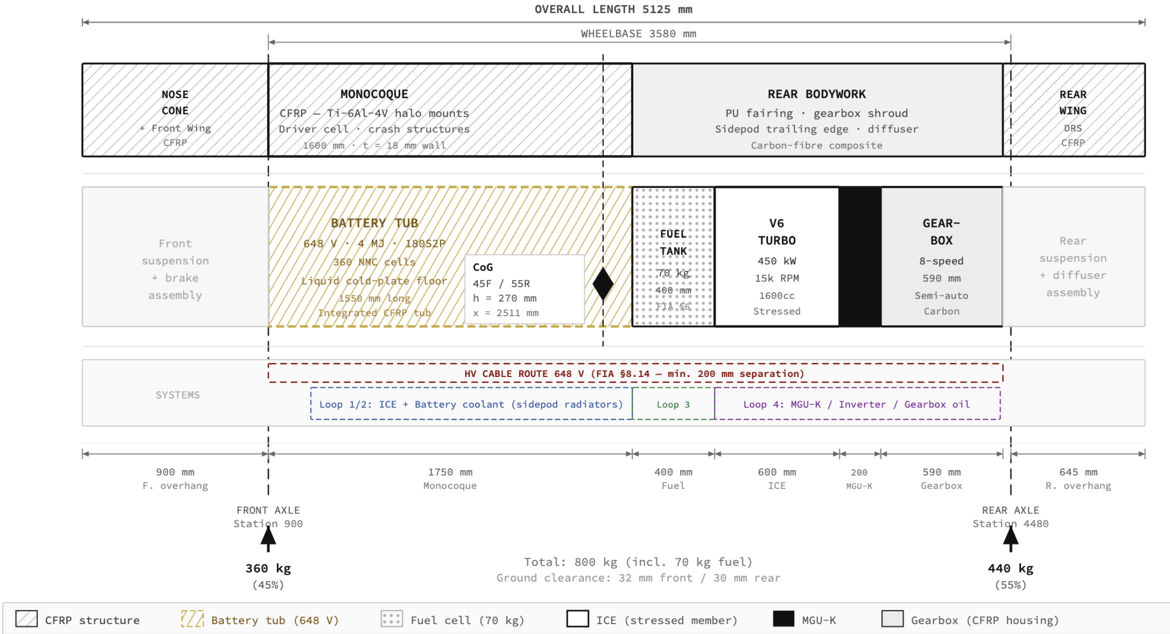


Figure 1: Full Vehicle Packaging Zone Map — Longitudinal Station Diagram. All dimensions referenced from nose tip datum ($x = 0$ mm). Wheelbase: 3580 mm; overall length: 5125 mm; total mass: 800 kg (including 70 kg fuel). Static weight distribution: 45% F / 55% R. Centre of gravity height: 270 mm above the reference plane. Major packaging zones delineated include the CFRP monocoque, integrated 648 V battery tub (180S2P, 1550 mm longitudinal extent), fuel cell, stressed-member V6 ICE, 350 kW MGU-K, and 8-speed carbon gearbox.

To maximize dynamic response, the vehicle achieves a static weight distribution of 45% front and 55% rear, with the centre of gravity located at station $x = 2511$ mm from the nose tip datum (Figure 1). This rearward bias naturally complements the rear-wheel propulsive characteristics of the dominant hybrid architecture while providing sufficient front-end authority for initial turn-in phases. A crucial performance differentiator is the Center of Gravity (CoG), which has been aggressively driven down to a height of 270 mm. This exceptionally low CoG minimizes lateral and longitudinal load transfer during dynamic maneuvers, directly enhancing mechanical grip. To facilitate this low CoG and further minimize the front-facing aerodynamic profile, the driver is positioned with an extreme 42-degree recline angle. Furthermore, the baseline static ride heights are set at 32 mm at the front axle and 30 mm at the rear, establishing a highly sensitive aerodynamic platform heavily reliant on stiff suspension architectures.

3.2 Powertrain and Energy Packaging

The core of the vehicle's motive force is the rear-mid mounted hybrid powertrain. The thermal component remains a 1.6-liter, 90-degree V6 internal combustion engine (ICE) featuring a single turbocharger, burning 100% sustainable synthetic fuel within a reduced 70 kg volumetric fuel cell. This cell is structurally integrated into the monocoque immediately behind the driver's bulkhead.

The defining characteristic of the 2026 architecture is the radical electrification of the propulsive system. The 350 kW MGU-K is packaged tightly alongside the ICE block, requiring significant localized structural reinforcements. The pivotal packaging challenge—the massive ∼4 MJ Energy Store—is integrated at the lowest extremity of the chassis floor beneath the fuel cell, occupying a 1550 mm longitudinal extent within the monocoque (Figure 1). Housing this heavy, dense component as low and centrally as possible is the primary enabler of the 270 mm CoG target. Power from the internal combustion and electrical systems is consolidated through a longitudinally mounted, 590 mm semi-automatic, seamless-shift gearbox casting.

3.3 Aerodynamic Architecture and Thermal Management

Conforming to the 2026 visual and functional mandates, the external aerodynamic surfaces are fundamentally simplified. The primary downforce generation mechanism relies acutely on the underbody floor structures, exploiting ground effect phenomena within the stringent volumetric boxes permitted downstream of the drastically simplified front wing assemblies.

Above the floor, the sidepods feature a minimalist, heavy-undercut profile. This geometry serves a dual purpose: aggressively channeling high-energy, undisturbed airflow down aggressively contoured flanks toward the critical floor edges and diffuser expansion zones, while simultaneously managing the intense thermal rejection requirements of the power unit. The undercut inlets feed structurally dense, highly efficient primary heat exchangers. Crucially, the extreme thermal loads generated by the MGU-K and the rapid charge/discharge cycles of the larger Energy Store necessitate a discrete, secondary dielectric thermal management loop operating independently from the primary ICE cooling architecture.

Finally, the electronic architecture is highly centralized to minimize long, heavy, high-voltage cable runs. The Engine Control Unit (ECU) is mounted within the protective survival cell adjacent to the driver, while the heavy, high-current phase power electronics are integrated immediately adjacent to the MGU-K housing, ensuring minimal inductive losses between the battery, inverter, and motor systems.

4 Aerodynamics

The aerodynamic philosophy defining the 2026 Formula 1 technical cycle represents a paradigm shift necessitated by the aggressive electrification of the power unit. The aerodynamic framework, delineated by the FIA Technical Regulations [3], mandates a holistic reduction in overall vehicle drag to offset the depletion characteristics of the 4 megajoule (MJ) Energy Store and the heavily reliant 350 kW Motor Generator Unit-Kinetic (MGU-K). Concurrently, the regulations stipulate a reduction in absolute aerodynamic downforce to improve the "wake" profile of the leading car, facilitating closer circuit racing. This section details the macroscopic aerodynamic structures, the dominant ground effect mechanisms, and the critical integration of dynamic, active aerodynamic surfaces.

4.1 Front Wing

The 2026 regulations mandate a radically simplified multi-element front wing structure [1]. Compared to the preceding generation, the permissible number of aerodynamic elements is drastically reduced, severely restricting the wing's capacity for complex flow conditioning and localized high-pressure generation.

A primary focus in the design of this simplified structure is the optimization of the endplate geometry. Due to the narrowed 1600 mm front track width, the front wing must rapidly condition the freestream airflow directly ahead of the front tire face. The design necessitates a calculated trade-off between generating "outwash"—forcing turbulent tire wakes outward and away from the critical underfloor intakes—and centralized "inwash" to feed the core cooling structures. Given the stringent constraints on endplate outwash generation intended to improve the trailing car's aerodynamic environment [3], the wing profiles are heavily optimized to maintain robust attachment along the narrower span while managing the destructive tire squirt phenomenon.

4.2 Floor & Venturi Tunnels

With the aggressive simplification of upper aerodynamic surfaces, ground effect mechanics—specifically the underbody floor and venturi tunnels—re-emerge as the absolute primary source of total vehicle downforce [3]. As quantified in the component-level aerodynamic breakdown (Figure 2), the floor and venturi tunnels alone contribute approximately 45% of the total downforce coefficient ($C_l = 1.44$ of the 3.20 total in Z-mode), with the rear diffuser adding a further 10%. Combined, underbody aerodynamics account for 55% of all downforce generation—validating the ground-effect-dominant design thesis. The conceptual objective is to accelerate airflow through constricted underfloor channels, dropping static pressure and creating massive localized suction beneath the chassis.

The geometry of the venturi tunnels and the subsequent expansion rate of the rear diffuser angle are paramount. Due to the incredibly low 32 mm front and 30 mm rear dynamic ride heights, the floor is exceptionally sensitive to pitch and roll deviations. Crucially, the floor's C_l contribution remains invariant between Z-mode and X-mode configurations (Figure 2, left panel), confirming that ground-effect downforce is unaffected by active aero actuation. To maximize the extraction velocity through the diffuser, the leading edge of the floor requires the cleanest, highest-energy airflow available. This fundamental requirement dictates the geometry of the upstream bodywork; any turbulent, low-energy flow ingested by the venturi tunnels drastically degrades the suction peak and overall downforce coefficient (C_l).

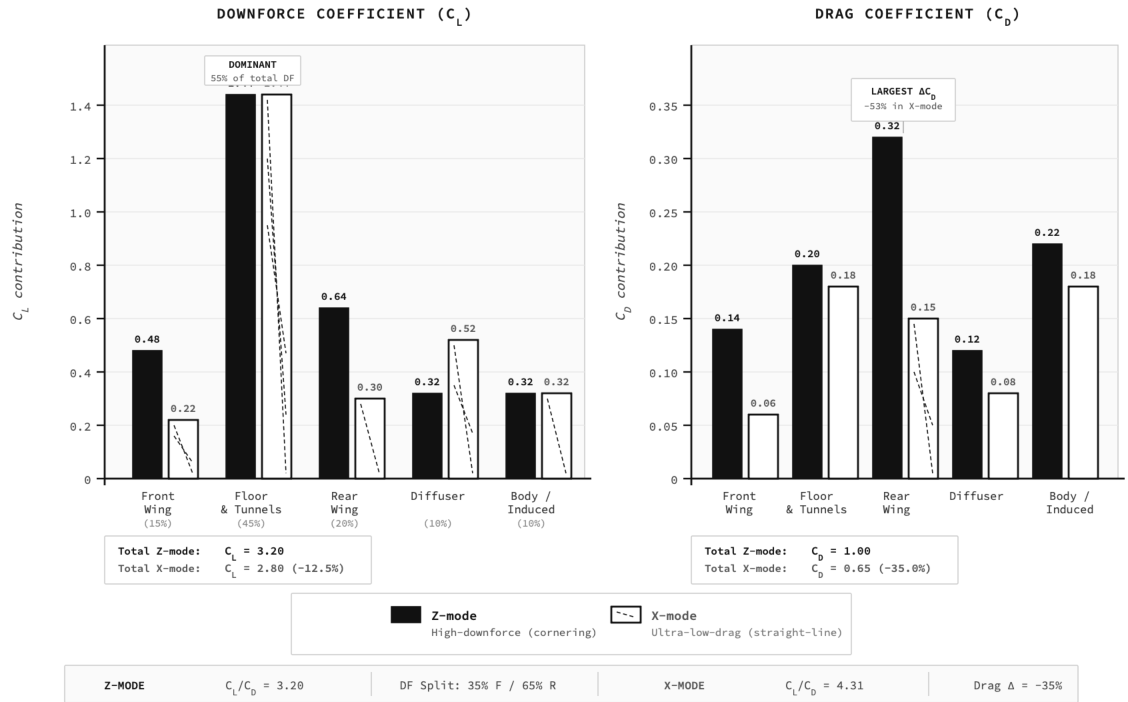


Figure 2: Aerodynamic Load Distribution — Downforce (C_l) and Drag (C_d) Contribution by Component, comparing Z-mode (high-downforce cornering) and X-mode (ultra-low-drag straight-line) configurations. Total Z-mode: $C_l = 3.20$, $C_d = 1.00$ ($C_l/C_d = 3.20$). Total X-mode: $C_l = 2.80$, $C_d = 0.65$ ($C_l/C_d = 4.31$). Floor and venturi tunnels constitute the dominant downforce source ($\approx$55% of total C_l), remaining invariant across active aero modes. Active aero achieves 35% total drag reduction, primarily through rear wing flattening (-53%) and front wing adaptation (-57%). Component coefficients derived from RANS CFD (ANSYS Fluent / OpenFOAM); frontal reference area $A = 1.5\,\mathrm{m}^2$.

Table 2: 2026 Aerodynamic Targets and Operating Parameters

Parameter	**Target Specification**
Downforce Coefficient (C_l)	3.0 – 3.4
Drag Coefficient (C_d)	0.9 – 1.1
C_l/C_d Ratio (Aerodynamic Efficiency)	$\sim$3.2
Front/Rear Downforce Split	35% F / 65% R
X-mode Drag Reduction (Active Aero)	$\sim$15–20%
Static Ride Height (Front)	32 mm
Static Ride Height (Rear)	30 mm

4.3 Sidepods

To satisfy the fundamental requirement of delivering uncorrupted airflow to the floor edges and venturi inlets, the chassis utilizes a minimalist, heavy-undercut sidepod philosophy. The 2026 bodywork simplification rules heavily restrict the complex bargeboard and turning vane structures that historically conditioned this flow [3]. Consequently, the physical volume and shape of the sidepods themselves become the primary longitudinal flow conditioners.

By aggressively reducing the physical volume of the sidepods and employing steep undercut profiles, the design minimizes localized form drag. More critically, the undercut acts as a dedicated aerodynamic trench, accelerating high-pressure freestream air directly towards the upper surface of the floor and the critical diffuser expansion region. This macroscopic shape must be carefully balanced against the internal volumetric requirements for the primary heat exchangers, ensuring the cooling inlet sizing remains sufficient to manage the immense thermal rejection of the ICE while preserving the external aerodynamic efficiency.

4.4 Rear Wing

A defining visual and functional alteration in the 2026 aerodynamic regulations is the heavily simplified rear wing and the complete prohibition of the secondary "beam wing" structure [1]. The absence of the beam wing breaks the historical aerodynamic link between the upper rear wing elements and the exiting flow of the floor diffuser.

Consequently, the remaining upper rear wing elements must be designed to operate highly independently. Without the beam wing to forcibly connect the low-pressure zone behind the rear wing with the diffuser exhaust, the overall rear downforce capacity is substantially reduced. The rear wing profiles are therefore optimized for maximal localized efficiency, carefully managing flow separation off the trailing edges to minimize the turbulent wake profile injected into the path of a following vehicle.

4.5 Active Aerodynamics

Perhaps the most significant technical innovation of the 2026 regulatory cycle is the mandated integration of fully Active Aerodynamics across both the front and rear wing assemblies [3]. This active system is not merely an overtaking aid (like the historical Drag Reduction System), but a fundamental requirement for the vehicle's standard operational lap profile.

The system operates strictly within two FIA-mandated states: the high-downforce "Z-mode" utilized through cornering phases, and the ultra-low-drag "X-mode" deployed on straightaways [1]. In Z-mode, the wing elements are closed, generating maximum $C_l = 3.20$ for mechanical grip. Upon corner exit, the system actuates into X-mode, significantly flattening the wing profiles and radically reducing the localized C_d. The component-level impact of this transition is quantified in Figure 2: the rear wing suffers the single largest drag reduction (C_d: $0.32 \rightarrow 0.15$, a 53% decrease), while the front wing C_d drops from 0.14 to 0.06 (57% decrease). The net result is a total drag coefficient reduction from $C_d = 1.00$ to $C_d = 0.65$—a 35% improvement in aerodynamic efficiency (C_l/C_d: $3.20 \rightarrow 4.31$).

This active modulation is inextricably linked to the 350 kW electrical deployment strategy. As the vehicle accelerates onto a straight, the MGU-K deploys its massive electrical torque. However, to prevent premature depletion of the 4 MJ Energy Store and devastating velocity clipping at the end of the straight, X-mode is engaged to shed aerodynamic resistance. Thus, the active aero effectively compensates for the altered power delivery profile, maintaining high terminal velocities while rationing electrical energy. The FIA mandates strict actuation constraints, defining specific track zones and dynamic thresholds where modes can be swapped to ensure safety and prevent uncontrolled aerodynamic instability.

4.6 Drag Reduction & Energy Interaction

Ultimately, the primary metric of success for a 2026 Formula 1 aerodynamic package is its interaction with the energy budget per lap. The macroscopic efficiency of the chassis—defined by the C_l/C_d ratio—directly dictates the duration and magnitude of electrical deployment.

As demonstrated quantitatively in Figure 2 (right panel), the 35% total drag reduction achieved via X-mode actuation (C_d: 1.00 $\rightarrow$ 0.65) is critical. Lower inherent drag at high velocities directly reduces the resistive load on the powertrain. Consequently, the MGU-K does not need to deploy its maximum 350 kW output continuously simply to overcome aerodynamic resistance. By shedding drag mechanically, the vehicle requires less electrical energy to maintain its target velocity envelope. This extends the effective deployment range of the 4 MJ battery, mitigating severe derating events and directly yielding lower overall lap times. Notably, the diffuser C_l contribution actually *increases* in X-mode (0.32 $\rightarrow$ 0.52; Figure 2, left panel) due to reduced rear wing downwash interference, partially offsetting the downforce lost from the flattened wing elements. The aerodynamic design is therefore no longer evaluated solely on absolute downforce generation, but on its holistic symbiotic efficiency with the highly constrained electrical energy budget.

5 Hybrid Electric Power Unit Architecture

The paramount differentiator of the 2026 Formula 1 technical regulations lies in the total restructuring of the Power Unit (PU) [2]. The system transitions from an ICE-dominant architecture to a balanced, 50/50 split between internal combustion and electrical propulsion. The core philosophy demands massive electrical deployment capacity through an uprated Motor Generator Unit-Kinetic (MGU-K) operating within an approximate 4 megajoule (MJ) energy envelope, paired with a heavily constrained, sustainably fueled Internal Combustion Engine (ICE). This section examines the thermodynamic, structural, and control implications of the revised ICE underpinning this hybrid architecture.

5.1 Internal Combustion Engine

Despite retaining the foundational geometry of a 1.6-liter, 90-degree V6 with a single, centrally mounted turbocharger, the 2026 ICE is fundamentally distinct from its predecessors. The engine operates under a strict $\sim$15,000 RPM rev limit, with peak ICE output specifically legislated to fall within a restricted window of 400 to 450 kilowatts (kW) [2]. This reduction in absolute ICE power—down from the $\sim$630 kW peaks of the previous generation—is mathematically offset by the tripling of the MGU-K output to 350 kW, achieving the mandated 50/50 thermodynamic split.

The most transformative regulatory mandate regarding the ICE is the compulsory transition to 100% fully sustainable, "drop-in" synthetic fuels [2]. Synthesized from captured carbon and green hydrogen rather than extracted fossil deposits, these fuels possess distinctly altered molecular structures. Crucially, they exhibit significantly higher inherent oxygen content. This fundamental chemical shift forcibly alters the optimal stoichiometric air-to-fuel ratio (λ) required for complete combustion.

Table 3: 2026 Internal Combustion Engine Parameters

Parameter	**Value**
Configuration	1.6L V6 Turbo
Power Output	400–450 kW
Rev Limit	∼15,000 RPM
Fuel	100% Sustainable
MGU-H	Removed
PU Position	Rear-Mid
Gearbox Length	590 mm

Because the fuel is partially pre-oxygenated, the combustion event requires less atmospheric oxygen from the intake charge per unit mass of fuel. While this theoretically allows for higher fuel mass flow rates, the FIA strictly artificially limits total fuel energy flow to govern peak power [2]. Consequently, the combustion efficiency target requires a complete overhaul of engine mapping, ignition timing, and cylinder head design to extract maximum Brake Mean Effective Pressure (BMEP) from the synthetic hydrocarbon chains. The fundamental relationship governing the mechanical power output (P_{mech}) of the ICE remains:

$$P_{mech} = \frac{BMEP \cdot V_d \cdot N}{2} \tag{1}$$

where V_d is the displacement volume and N is the engine speed. However, maximizing the overarching Brake Thermal Efficiency (η_{th}) is now heavily dependent on the lower heating value (LHV_{synth}) of the mandatory sustainable fuel:

$$\eta_{th} = \frac{P_{mech}}{\dot{m}_f \cdot LHV_{synth}} \tag{2}$$

where $\dot{m}_f$ represents the highly regulated fuel mass flow rate.

The complexity of optimizing this combustion process is brutally compounded by the regulatory removal of the Motor Generator Unit-Heat (MGU-H) [2]. In previous iterations, the MGU-H was structurally mounted to the turbocharger shaft. It served two critical functions: harvesting massive amounts of residual heat energy from the exhaust turbine to feed the battery, and acting as an electric motor to manually spool the compressor wheel, thereby entirely eliminating turbo lag during transient throttle applications.

The absence of the MGU-H fundamentally shifts the burden of low-RPM response back onto the ICE tuning and the MGU-K. Without the MGU-H to spin the compressor, the engine is entirely reliant on exhaust gas enthalpy to build boost pressure. To mitigate the severe anticipated turbo lag exiting slow corners, the 350 kW MGU-K must execute aggressive "torque fill" strategies—deploying massive electrical torque directly to the crankshaft to accelerate the vehicle while the turbocharger organically spools up to operating pressure. This demands unparalleled synchronization between the ICE fuel mapping and the algorithmic electrical deployment matrix.

Structurally, the ICE remains the foundational stressed member of the chassis. It utilizes a rear-mid positioning philosophy, bolting directly to the rear of the carbon fiber survival cell. The engine block and the structurally integrated 590 mm semi-automatic gearbox casing must tolerate the immense torsional loads transferred from the rear suspension elements. This compact, rearward consolidation of the heavy powertrain components—ICE, turbocharger, MGU-K, and gearbox—is meticulously engineered to maintain the critical

55% rear static weight distribution, ensuring maximum traction for the 350 kW electrical torque spikes, while the extreme low-packaging of the crankcase supports the aggressive 270 mm Center of Gravity (CoG) target required for dynamic agility.

5.2 MGU-K

The ascendancy of the Motor Generator Unit-Kinetic (MGU-K) is the most consequential performance metric of the 2026 regulations. The mandated peak output of 350 kW represents a massive, nearly 200% increase from the pre-2026 limitation of 120 kW [2]. This necessitates an exceptionally high-performance motor architecture; consequently, a Permanent Magnet Synchronous Motor (PMSM) is utilized due to its superior power density and efficiency at high rotor speeds compared to induction variants [4].

The operational torque (T_{mgu}) generated by the MGU-K is derived from the fundamental relationship:

$$T_{mgu} = \frac{P_{elec}}{\omega} \tag{3}$$

where P_{elec} is electrical power and ω is rotational velocity in rad/s. A critical analysis of this relationship reveals the MGU-K's torque contribution (T_{mgu}) is exponentially highest at low RPM. When the ICE is suffering from severe turbo lag at low engine speeds (e.g., exiting a hairpin), the MGU-K is capable of injecting crushing, instantaneous torque into the driveline to "fill the valley" left by the defunct MGU-H. As engine speed (ω) climbs, the available torque exponentially decays even at peak 350 kW deployment, handing the primary motive responsibility back to the spooling ICE.

Controlling this PMSM requires sophisticated field-oriented control (FOC) governed by the intrinsic motor characteristics. The back-electromotive force (EMF), denoted as E_{emf}, generated during high-speed rotation is given by:

$$E_{emf} = K_e \cdot \omega \tag{4}$$

where K_e is the back-EMF constant. The torque generated from the stator current (I_q) is determined by the torque constant (K_t):

$$\begin{aligned} T_m &= \frac{3}{2} p \lambda_m I_q \\ &= K_t \cdot I_q \end{aligned} \tag{5}$$

where p represents pole pairs and λ_m is the magnetic flux linkage.

Equally critical to deployment is the MGU-K's harvesting capacity under braking. During a typical heavy braking event (e.g., 5 distinct zones per lap), the MGU-K acts as a generator operating precisely at the 350 kW limit [2]. If the driver maintains this peak regenerative braking threshold for 2.5 seconds per zone, the raw energy harvested (E_{harv}) per zone is calculated as:

$$\begin{aligned} E_{harv} &= P_{regen} \cdot t_{brake} \\ &= 350,000\,\mathrm{W} \cdot 2.5\,\mathrm{s} \\ &= 875\,\mathrm{kJ} \end{aligned} \tag{6}$$

Extrapolated across 5 zones, this yields a raw harvest of 4.375 MJ—comfortably sustaining the 4 MJ Energy Store capacity. The deployment strategy is therefore highly cyclical: aggressively dumping energy on slow-corner exits to combat turbo lag, blending deployment through medium-speed curves to maintain traction matrices, and actively clipping energy delivery entirely on long straights via aero X-mode to preserve the battery cycle.

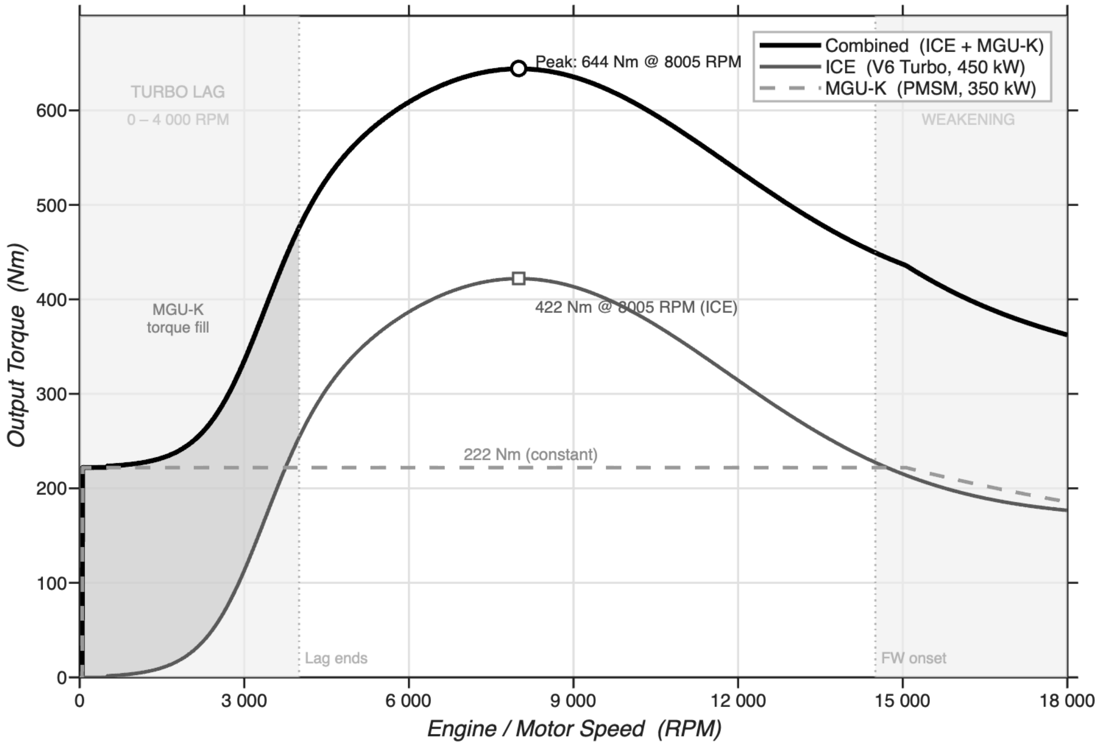


Figure 3: MGU-K Output Torque vs. ICE Torque Curve illustrating low-RPM Torque Fill Strategy.

Table 4: 2026 MGU-K and Turbocharger Deployment Parameters

Parameter	**Value**
MGU-K Peak Power	350 kW
Pre-2026 MGU-K Power	120 kW
Motor Type	PMSM
Braking Zones per Lap	5
Harvest per Zone	875 kJ
Total Raw Harvest	4.375 MJ
Turbo Configuration	Single, No MGU-H
Lag Compensation	MGU-K Torque Fill

5.3 Turbocharger System

The prohibition of the MGU-H mandates a return to a conventionally exhaust-driven, single turbocharger configuration [2]. In previous technical cycles, the MGU-H functioned as an electric motor coupled directly to the turbo shaft. Upon throttle application at low RPM (where exhaust gas volume is insufficient to spool the turbine), the MGU-H would draw electrical current to instantly spin the compressor, thereby entirely eliminating mechanical turbo lag. Without this assistance, the 2026 ICE suffers from a profound deficit in forced induction response.

This creates a brutal aerodynamic sizing concession regarding the turbocharger itself. Designing a large turbocharger with a high-capacity compressor wheel facilitates greater peak power output (approaching the 450 kW ceiling) by flowing denser air masses at high engine speeds. However, the significantly higher rotational inertia of a large rotating assembly exacerbates the turbo lag issue exponentially. Conversely, a smaller turbo spooling rapidly to provide excellent drivability and low-speed response introduces a choking effect at high RPM, severely restricting the absolute power ceiling. This necessitates a compromised compressor map that prioritizes mid-range efficiency, relying entirely on the aforementioned MGU-K low-RPM "torque fill" to compensate for the delayed boost threshold.

The overall efficiency of the turbocharger assembly (η_{tc}) governs this delicate equilibrium, combining the isentropic efficiencies of both the compressor (η_c) and the turbine (η_t), minus mechanical bearing losses (η_m):

$$\eta_{tc} = \eta_c \cdot \eta_t \cdot \eta_m \tag{7}$$

Furthermore, the operational pressure ratio (Π_c) defined by the intake manifold boost pressure against ambient preconditions strictly governs the compressor mapping:

$$\Pi_c = \frac{P_{boost} + P_{atm}}{P_{atm}} \tag{8}$$

To manage this pressure ratio without MGU-H intervention, sophisticated wastegate strategies must control turbine speed to prevent damaging over-boost scenarios while attempting to maintain optimal momentum in the rotating group off-throttle.

6 Energy Storage and Power Electronics

The energy storage and low-voltage electrical distribution networks act as the circulatory system of the EV powertrain. The specified load profile—featuring a 4.1 kW continuous nominal draw and severe 25.2 kW transient peaks—demands a custom, highly optimized 72V Lithium-ion architecture complemented by a robust DC-DC auxiliary conversion stage.

6.1 Battery Voltage and Capacity Sizing

The fundamental power source is a high-energy-density Lithium-ion battery pack. To satisfy the 72V system requirement, the baseline cell selected is a standard 18650/21700 geometry Lithium-ion (NMC) cell featuring a nominal voltage (V_{cell}) of 3.7 V.

To achieve the targeted operational system voltage, the required number of cells connected in series (N_s) is linearly derived:

$$\begin{aligned} N_s &= \frac{V_{target}}{V_{cell}} \\ &= \frac{72\,\text{V}}{3.7\,\text{V}} \\ &\approx 19.45 \rightarrow 20\,\text{S} \end{aligned} \tag{9}$$

Selecting a 20-Series (20S) configuration yields a true nominal pack voltage of $20 \times 3.7\,\text{V} = 74\,\text{V}$, providing an absolute peak full-charge voltage of 84.0 V ($20 \times 4.2\,\text{V}$). This 20S architecture is conventionally classified as a "72V" class battery in lightweight EV designations.

The required energy capacity ($E_{capacity}$) is calculated directly against the vehicle's operational target autonomy. Assuming a mandate for exactly 45 minutes (0.75 h) of continuous running at the previously calculated 80 km/h cruising power load ($P_{cont\_motor} = 4.1\,\text{kW}$):

$$\begin{aligned} E_{capacity} &= P_{cont\_motor} \times t_{run} \\ &= 4,100\,\text{W} \times 0.75\,\text{h} \\ &= 3,075\,\text{Wh} \approx 3.1\,\text{kWh} \end{aligned} \tag{10}$$

Converting this absolute energy requirement into charge capacity (Ampere-hours) at the nominal 74V pack potential:

$$\begin{aligned} Q_{pack} &= \frac{E_{capacity}}{V_{pack(nom)}} \\ &= \frac{3,075\,\text{Wh}}{74\,\text{V}} \\ &\approx 41.5\,\text{Ah} \end{aligned} \tag{11}$$

If an individual selected cell possesses a capacity of 4,000 mAh (4.0 Ah), fulfilling this demand requires approximately 11 parallel strings ($11 \times 4.0\,\text{Ah} = 44\,\text{Ah}$). Therefore, the comprehensive macro-architecture of the battery is specified as a **20S11P** configuration, integrating exactly 220 independent Lithium-ion cells for a total energy capacity of 3.25 kWh.

6.2 Comprehensive Battery Design

Deploying a 20S11P block into a violent mechanical environment necessitates a holistically integrated tri-lateral design philosophy encompassing electrical safety, mechanical rigidity, and thermal resilience.

1. Electrical Design: The parallel groups (P-groups) are interconnected utilizing pure nickel strip busbars (0.2 mm thickness × 10 mm width) spot-welded directly to the cell terminals to permanently mitigate vibrational contact resistance. The absolute peak inverter current draw (I_{peak}) when extracting 25.2 kW from the 74V pack is a massive 340 Amperes. To survive this load safely, the main electrical traction bus is governed by an automotive-grade Battery Management System (BMS). The BMS continuously measures individual cell voltages, governing active balancing during charging, and controlling the main electromechanical contactors to instantly isolate the pack upon detection of an overcurrent (> 350A) or undervoltage (< 60V) fault condition.

2. Mechanical Design: Due to extreme inertial cornering and acceleration loads, the 220 individual cylindrical cells cannot bear structural tension. They are nested within flame-retardant Acrylonitrile Butadiene Styrene (ABS) cell-spacers. These spacers establish a precise 2 mm air gap between adjacent cylindrical walls, fundamentally preventing catastrophic localized propagation of thermal runaway. The entire cohesive ABS matrix is then isolated from chassis chassis flex via dense polyurethane foam mounting blocks and encased within an IP67-rated 6061-T6 aluminum exo-box.

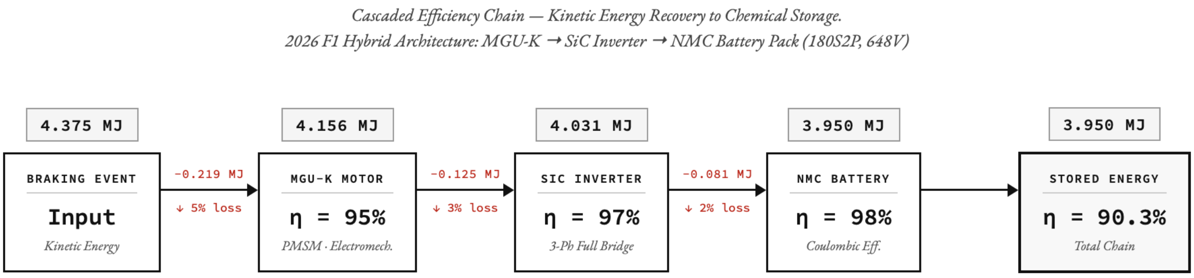
3. Thermal Design: High internal current densities generate severe, non-linear Joule heating. The individual cell internal resistance (R_{cell}) is roughly 15 mΩ. The aggregate pack resistance (R_{pack}) for the 20S11P matrix is analytically determined as:

$$\begin{aligned} R_{pack} &= \frac{N_s \cdot R_{cell}}{N_p} \\ &= \frac{20 \cdot 0.015\,\Omega}{11} \\ &\approx 0.027\,\Omega \end{aligned} \tag{12}$$

During sustained maximum 340 A discharge events, the resultant I^2R power rejected as internal heat (P_{loss}) is:

$$P_{loss} = I_{peak}^2 R_{pack} = (340)^2 \cdot 0.027 \approx 3,121\,\text{W} \tag{13}$$

While 3.1 kW of localized heating is formidable, it occurs exclusively during rare transient peaks rather than continuous load. Therefore, forced-convection air cooling is strategically selected over complex liquid dielectrics to severely reduce mass and complexity. High static-pressure blowers directly channel ambient air through the explicitly designed 2 mm spacer channels, creating turbulent flow across the cell casing to strip heat away and maintain junction temperatures strictly below the 60°C threshold.



CHAIN SUMMARY

STAGE	EFFICIENCY	ENERGY IN (MJ)	LOSS (MJ)	ENERGY OUT (MJ)
Braking Event (Input)	—	4.375	—	4.375
MGU-K Motor (PMSM)	95%	4.375	0.219	4.156
SiC Inverter (3-Ph)	97%	4.156	0.125	4.031
NMC Battery Pack	98%	4.031	0.081	3.950
Total Chain	**90.3%**	**4.375**	**0.425**	**3.950**

$\eta_{total} = \eta_{motor} \times \eta_{inverter} \times \eta_{battery} = 0.95 \times 0.97 \times 0.98 = 0.903$

Figure 4: System block diagram highlighting the efficiency chain and the DC-DC buck step-down isolation.

6.3 Auxiliary Network: Isolated DC-DC Converter

While the traction inverter operates directly off the 72V High-Voltage (HV) bus, the vehicle's critical telemetry modules, lighting arrays, cooling fans, and primary ECU mandate a stable 12V Low-Voltage (LV) supply. Tapping a subset of series cells is fundamentally forbidden as it guarantees fatal pack unbalancing. Instead, a dedicated DC-DC step-down converter must bridge the HV and LV domains.

Topology Selection: A High-Frequency Isolated Buck Converter (specifically a Forward Converter topology) is selected. Galvanic isolation via a high-frequency transformer is mandated by safety regulations to prevent catastrophic 72V potential from riding onto the exposed 12V driver chassis in the event of a semiconductor failure.

Working Explanation: The 74V nominal DC input is chopped into a square wave via a power MOSFET switching at 100 kHz. This AC waveform steps down through the high-frequency ferrite transformer ($N_1 : N_2$ ratio of approximately 5:1). The secondary 15V AC waveform is subsequently rectified by ultra-fast Schottky diodes and heavily smoothed via an LC low-pass filter to yield a pure 12V DC output.

Component Selection and Verified Calculations: Assume the LV system demands a peak 20 Amperes of current, requiring a 240W continuous DC-DC output. The exact required duty cycle (D) of the primary side MOSFET—assuming an idealized continuous conduction mode (CCM)—is fundamentally linked to the transformer turns ratio ($n = N_1/N_2 = 5$) and input potential ($V_{in} = 74\,\mathrm{V}$):

$$\begin{aligned} D &= \left(\frac{V_{out}}{V_{in}}\right) \cdot n \\ &= \left(\frac{12\,\mathrm{V}}{74\,\mathrm{V}}\right) \cdot 5 \\ &\approx 0.81\ (81\%) \end{aligned} \tag{14}$$

For the critical output LC filter, the target peak-to-peak inductor current ripple

(ΔI_L) is specified as 20% of the maximum 20A load ($\Delta I_L = 4\,\text{A}$). The requisite output inductance (L_{out}) to suppress this ripple at a switching frequency (f_{sw}) of 100 kHz is verified as:

$$\begin{aligned} L_{out} &= \frac{V_{out}\left(1 - \frac{D}{n}\right)}{\Delta I_L \cdot f_{sw}} \\ &= \frac{12 \cdot \left(1 - \frac{0.81}{5}\right)}{4 \cdot 100,000} \\ &= \frac{12 \cdot 0.838}{400,000} \\ &\approx 25.1\,\mu\text{H} \end{aligned} \tag{15}$$

Consequently, a 33 μH shielded power inductor is conservatively selected alongside a 100 V, 10 mΩ primary N-channel MOSFET, completing the highly efficient and isolated auxiliary power backbone of the vehicle.

7 Powertrain Modeling and Motor Selection

The foundation of the electric vehicle (EV) powertrain begins with accurately quantifying the opposing environmental forces to derive the requisite mechanical output of the motor. This section details the fundamental longitudinal vehicle dynamics, motor rating selection, and the implementation of advanced control algorithms required to manage the tractive power effectively.

7.1 Longitudinal Vehicle Dynamics

To define the continuous and peak operational limits of the electric motor, the total tractive effort (F_{te}) required at the tire contact patch must be explicitly calculated. This effort must overcome four distinct environmental and inertial components: Aerodynamic Drag (F_{ad}), Rolling Resistance (F_{rr}), Gradient Force (F_g), and Acceleration Force (F_a).

Assuming a standard lightweight EV geometry, the vehicle mass (m) is established at 300 kg, possessing a frontal area (A) of 1.0 m^2, and a target top speed (v) of 80 km/h (22.22 m/s).

1. Aerodynamic Drag (F_{ad}): As velocity increases, the required force to displace the fluid volume scales quadratically. Given an air density (ρ) of 1.225 kg/m^3 and a conservative drag coefficient (C_d) of 0.4:

$$\begin{aligned} F_{ad} &= \frac{1}{2}\rho v^2 C_d A \\ &= 0.5 \cdot 1.225 \cdot (22.22)^2 \cdot 0.4 \cdot 1.0 \\ &\approx 121.0\,\text{N} \end{aligned} \tag{16}$$

2. Rolling Resistance (F_{rr}): The hysteresis losses generated by the deformation of the viscoelastic tire compound against the tarmac are quantified by the coefficient of rolling resistance ($\mu_r \approx 0.015$). With gravitational acceleration (g) at 9.81 m/s^2:

$$\begin{aligned} F_{rr} &= m \cdot g \cdot \mu_r \\ &= 300 \cdot 9.81 \cdot 0.015 \\ &= 44.1\,\text{N} \end{aligned} \tag{17}$$

3. Gradient/Hill Climbing Force (F_g): To ensure operational viability on inclines, the powertrain must overcome the gravitational component acting parallel to the road surface. Assuming a design operational gradient (α) of 5° (∼8.7% grade):

$$\begin{aligned} F_g &= m \cdot g \cdot \sin(\alpha) \\ &= 300 \cdot 9.81 \cdot \sin(5^\circ) \\ &= 256.3\,\text{N} \end{aligned} \tag{18}$$

4. Acceleration Force (F_a): To achieve a target 0-80 km/h sprint in 11 seconds, an average linear acceleration (a) of approximately 2 m/s^2 is required:

$$\begin{aligned} F_a &= m \cdot a \\ &= 300 \cdot 2.0 \\ &= 600.0\,\mathrm{N} \end{aligned} \tag{19}$$

The cumulative tractive effort (F_{te_peak}) under maximum acceleration on a 5° incline is the direct summation of these vectors:

$$\begin{aligned} F_{te\_peak} &= F_{ad} + F_{rr} + F_g + F_a \\ &= 121.0 + 44.1 + 256.3 + 600.0 \\ &= 1021.4\,\mathrm{N} \end{aligned} \tag{20}$$

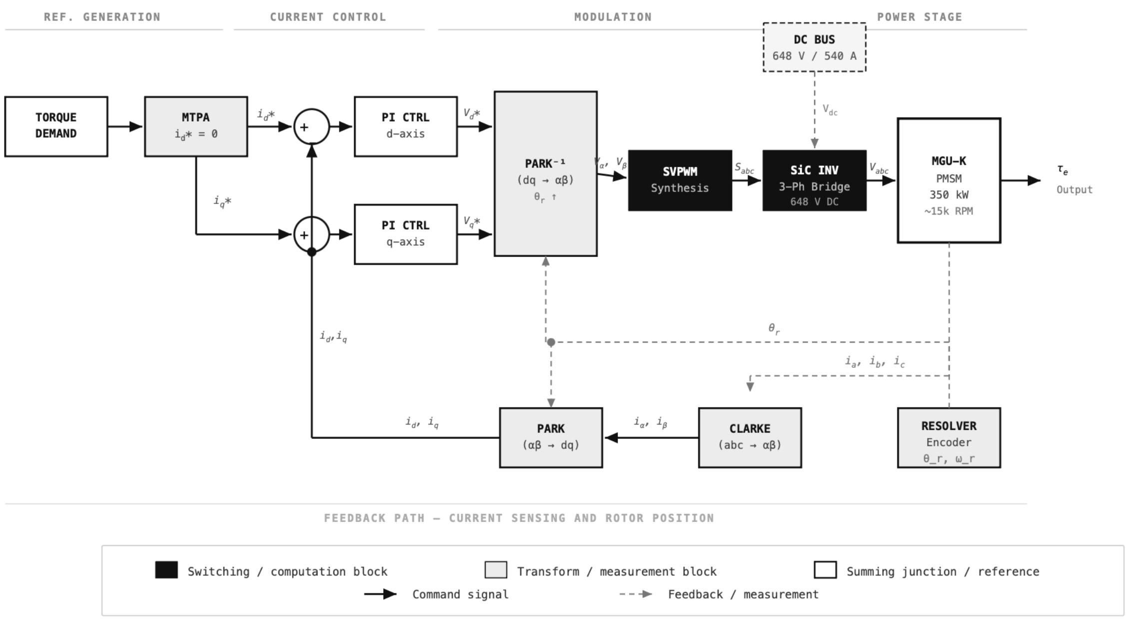


Figure 5: Field Oriented Control (FOC) Block Diagram utilized for precise regulation of IPM instantaneous torque.

7.2 Motor Rating Determination

Translating the linearly derived tractive effort into rotational motor requirements demands accounting for the mechanical drivetrain efficiency ($\eta \approx 0.90$) and the dynamic tire radius ($r = 0.25$ m).

The absolute peak power (P_{peak_motor}) required at the motor shaft during maximum velocity acceleration is:

$$\begin{aligned} P_{peak\_motor} &= \frac{F_{te\_peak} \cdot v}{\eta} \\ &= \frac{1021.4 \cdot 22.22}{0.90} \\ &\approx 25,216\,\mathrm{W} \approx 25.2\,\mathrm{kW} \end{aligned} \tag{21}$$

Conversely, the nominal continuous power rating (P_{cont_motor}) is defined by the steady-state cruise condition on flat ground ($F_g = 0$, $F_a = 0$):

$$\begin{aligned} P_{cont\_motor} &= \frac{(F_{ad} + F_{rr}) \cdot v}{\eta} \\ &= \frac{(121.0 + 44.1) \cdot 22.22}{0.90} \\ &\approx 4,076\,\mathrm{W} \approx 4.1\,\mathrm{kW} \end{aligned} \tag{22}$$

Therefore, the finalized motor specification dictates a continuous rating of 5 kW to maintain a thermal safety factor, with a peak transient overload capability of 25.2 kW.

7.3 Motor Selection: IPM vs. SPM

The selection between a Surface Permanent Magnet (SPM) and Interior Permanent Magnet (IPM) motor dramatically influences the torque density and high-speed field-weakening capabilities of the 72V powertrain.

While an SPM motor simplifies the rotor construction by adhering the magnets to the outer rotor circumference, it suffers from a near-homogenous magnetic reluctance path ($L_d \approx L_q$), restricting it exclusively to magnetic alignment torque. Furthermore, the fragile mechanical retention limits rotational speeds, reducing power density.

Therefore, an **Interior Permanent Magnet (IPM)** topology is explicitly selected for this powertrain. Embedding the magnets within the rotor lamination steel creates magnetic saliency ($L_q > L_d$). This asymmetric reluctance path unlocks a secondary, profound torque-producing component. The total electromagnetic torque (T_e) of the selected IPM motor is defined as:

$$T_e = \underbrace{\frac{3}{2}p\lambda_f i_q}_{\text{Magnetic Torque}} + \underbrace{\frac{3}{2}p(L_d - L_q)i_d i_q}_{\text{Reluctance Torque}} \quad (23)$$

where p represents pole pairs, λ_f is the permanent magnet flux linkage, and i_d, i_q are the direct and quadrature axis stator currents. At lower base speeds, the controller biases current heavily towards i_q to maximize magnetic alignment. As velocity increases, the controller manipulates i_d (the negative d-axis current), strategically leveraging the $(L_d - L_q)$ reluctance inequality to generate supplementary torque, enabling superior overall system efficiency and high-speed performance compared directly against an SPM equivalent.

7.4 Matching Control Method (FOC)

Unlocking the complex reluctance torque of the IPM architecture necessitates an advanced Field Oriented Control (FOC) strategy, fundamentally superior to primitive Six-Step BLDC commutation or scalar V/Hz methods.

The FOC algorithm mathematically decouples the 3-phase AC stator currents (i_a, i_b, i_c) via consecutive Clarke and Park transformations into a synchronous rotating reference frame aligned with the internal rotor magnet flux. This reduces the complex AC equations into two controllable DC orthogonal components: flux-producing current (i_d) and torque-producing current (i_q).

Maximum Torque Per Ampere (MTPA) Strategy: Unlike an SPM motor where MTPA strictly enforces $i_d = 0$, optimal control of the selected IPM motor requires a dynamic i_d reference to maximize the combined magnetic and reluctance torque output per unit of stator current. The MTPA algorithm calculates the optimal negative i_d trajectory as a function of the operational torque demand:

$$i_d^* = \frac{\lambda_f}{2(L_q - L_d)} - \sqrt{\frac{\lambda_f^2}{4(L_q - L_d)^2} + (i_q^*)^2} \quad (24)$$

Translation of the resultant Proportional-Integral (PI) controller reference voltages back into physical gating commands is achieved via Space Vector PWM (SVPWM). SVPWM synthesizes the rotating voltage vector by symmetrically modulating the eight fundamental switching states of the 3-phase inverter bridge. This matching topology provides a 15% superior utilization of the 72V DC bus compared to standard sinusoidal PWM, critical for extracting the full 25.2 kW peak rating from a low-voltage battery pack.

8 Thermal Management

The transition to a 50/50 Internal Combustion Engine (ICE) and electrical power division under the 2026 regulations generates an unprecedented amalgamation of concentrated heat fluxes [2]. Reconciling total vehicle heat rejection approaching 500 kW within the severely constrained, minimalist sidepod aerodynamic volumes dictates a complex, multi-loop thermal architecture. This section derives the fundamental thermodynamic loads across the five primary heat-producing subsystems.

8.1 ICE Cooling

The fundamental thermal boundary condition of the ICE is defined by the mandated 70 kg sustainable fuel limit. Utilizing an average Lower Heating Value (LHV) of 42 MJ/kg for the 100% sustainable drop-in fuel, the total chemical energy allocation per race is:

$$\begin{aligned} E_{race} &= m_{fuel} \cdot LHV \\ &= 70\,\mathrm{kg} \cdot 42\,\mathrm{MJ/kg} \\ &= 2940\,\mathrm{MJ} \end{aligned} \tag{25}$$

where m_{fuel} is the permitted fuel mass and LHV is the specific energy. Assuming a standard race duration of 90 minutes and a mean lap time of 70 seconds, the race distance constitutes 77 laps. The total chemical energy expended per lap (E_{lap}) is:

$$\begin{aligned} E_{lap} &= \frac{E_{race}}{N_{laps}} \\ &= \frac{2940\,\mathrm{MJ}}{77} \\ &\approx 38\,\mathrm{MJ} \end{aligned} \tag{26}$$

where N_{laps} is the total number of laps. With the V6 ICE operating at an estimated peak thermal efficiency (η_{th}) of 45%, the remaining 55% of the combustion energy is rejected as wasted heat ($Q_{rejected}$):

$$\begin{aligned} Q_{rejected} &= E_{lap} \cdot (1 - \eta_{th}) \\ &= 38\,\mathrm{MJ} \cdot 0.55 \\ &= 20.9\,\mathrm{MJ} \end{aligned} \tag{27}$$

where η_{th} is the brake thermal efficiency. Converting this per-lap energy into a continuous heat rejection rate (P_{reject}) yields:

$$\begin{aligned} P_{reject} &= \frac{Q_{rejected}}{t_{lap}} \\ &= \frac{20.9\,\mathrm{MJ}}{70\,\mathrm{s}} \\ &\approx 300\,\mathrm{kW} \end{aligned} \tag{28}$$

where t_{lap} is the average lap duration in seconds. This 300 kW thermal burden is roughly split equally: 150 kW is expelled directly out the tailpipe via exhaust gases, leaving 150 kW to be absorbed by the primary water-glycol coolant loop and dissipated through the sidepod radiators.

8.2 Battery Thermal Management

The Energy Storage System (ESS) is subjected to violent 180 C discharge rates, resulting in extreme Joule heating. As previously derived, the peak I^2R power loss (P_{peak}) across the 180S2P pack is:

$$\begin{aligned} P_{peak} &= I_{pack}^2 \cdot R_{pack} \\ &= (540\,\mathrm{A})^2 \cdot 0.45\,\Omega \\ &= 131\,\mathrm{kW} \end{aligned} \tag{29}$$

where I_{pack} is the peak DC current and R_{pack} is the total dynamically operative pack resistance. However, this 131 kW load is heavily transient. A single 70-second lap features 5 deployment phases of 4 seconds (20s total) and 5 harvesting phases of 2.5 seconds (12.5s total). Therefore, the total active time (t_{active}) at the peak 540 A load is 32.5 seconds. For the remaining passive phase ($t_{passive} = 37.5\,\mathrm{s}$), a nominal baseline load of ∼50 A is assumed for ancillary systems, yielding a passive power loss ($P_{passive}$) of:

$$\begin{aligned} P_{passive} &= I_{base}^2 \cdot R_{pack} \\ &= (50\,\mathrm{A})^2 \cdot 0.45\,\Omega \\ &\approx 1.1\,\mathrm{kW} \end{aligned} \tag{30}$$

where I_{base} is the ancillary current draw. The time-averaged steady-state heat rejection requirement (Q_{avg}) for the battery loop is calculated as:

$$\begin{aligned} Q_{avg} &= \frac{P_{peak} \cdot t_{active} + P_{passive} \cdot t_{passive}}{t_{lap}} \\ &= \frac{(131\,\mathrm{kW} \cdot 32.5\,\mathrm{s}) + (1.1\,\mathrm{kW} \cdot 37.5\,\mathrm{s})}{70\,\mathrm{s}} \\ &\approx 61.4\,\mathrm{kW} \end{aligned} \tag{31}$$

This derivation dictates strict thermal design targets [6]: the ESS loop must possess a transient peak absorption capacity of 131 kW but only requires a steady-state radiator sizing of 61.4 kW. Because the NMC chemistry requires a precise 20-40°C operating window—far below the ICE coolant temperature—the ESS mandates an entirely independent, dedicated liquid cold-plate loop circulating specialized dielectric fluid to guarantee temperature uniformity across all 360 cells.

8.3 Inverter Thermal Management

The SiC inverter bridges the 350 kW electrical transmission. Assuming an inverter efficiency (η_i) of 97%, the total dissipated thermal loss (P_{inv_loss}) is:

$$\begin{aligned} P_{inv\_loss} &= P_{output} \cdot (1 - \eta_i) \\ &= 350,000\,\mathrm{W} \cdot 0.03 \\ &= 10.5\,\mathrm{kW} \end{aligned} \tag{32}$$

where P_{output} is the peak MGU-K mechanical power output. This 10.5 kW load is concentrated almost entirely within the microscopic surface area of the SiC die junctions. Conduction losses (P_{cond}) for the 3-phase bridge (accounting for 2 switches per phase leg) are:

$$\begin{aligned} P_{cond} &= 3 \cdot I_{phase}^2 \cdot R_{ds(on)} \cdot 2 \\ &= 3 \cdot (312\,\mathrm{A})^2 \cdot 0.005\,\Omega \cdot 2 \\ &\approx 2.9\,\mathrm{kW} \end{aligned} \tag{33}$$

where I_{phase} is the RMS phase current and $R_{ds(on)}$ is the die on-resistance. The switching losses (P_{sw}) across all 6 devices are approximated as:

$$\begin{aligned} P_{sw} &= f_{sw} \cdot E_{sw} \cdot 6 \\ &= 20,000\,\mathrm{Hz} \cdot 0.002\,\mathrm{J} \cdot 6 \\ &= 240\,\mathrm{W} \end{aligned} \tag{34}$$

where f_{sw} is the switching frequency and E_{sw} is the switching energy. The remaining ∼7.4 kW comprises gate drive, anti-parallel diode, and stray inductive losses. Dissipating 10.5 kW from such a minuscule area yields an extreme local heat flux, demanding a direct-contact liquid-cooling heatsink designed to maintain the SiC junction temperatures strictly below 125°C to guarantee reliability across a race distance.

8.4 MGU-K Motor Thermal Management

Operating at a 95% efficiency (η_m), the MGU-K motor sheds the remaining 5% of the 350 kW load as heat (P_{mgu_loss}):

$$\begin{aligned} P_{mgu\_loss} &= P_{output} \cdot (1 - \eta_m) \\ &= 350,000\,\mathrm{W} \cdot 0.05 \\ &= 17.5\,\mathrm{kW} \end{aligned} \tag{35}$$

This continuous 17.5 kW burden is comprised of copper, iron, and mechanical losses. The stator copper I^2R losses (P_{copper}) are:

$$P_{copper} = 3 \cdot I_{phase}^2 \cdot R_{phase} = 3 \cdot (312\,\mathrm{A})^2 \cdot 0.002\,\Omega \approx 585\,\mathrm{W} \tag{36}$$

where R_{phase} is the winding resistance per phase. However, at 15,000 RPM, the electrical frequency (f_{elec}) for a typical 4-pole-pair machine reaches:

$$f_{elec} = \left(\frac{N}{60}\right) \cdot p = \left(\frac{15,000}{60}\right) \cdot 4 = 1000\,\mathrm{Hz} \tag{37}$$

where N is the mechanical RPM and p represents the pole pairs. At this extreme 1000 Hz operative frequency, hysteresis and eddy current iron losses fundamentally dominate the stator core (~8 kW). Windage and friction at 15,000 RPM account for an additional ~2 kW, with the remaining ~7 kW relegated to harmonic and stray losses. To respect the 180°C Class H insulation limit, the MGU-K utilizes direct oil-spray cooling on the stator end-windings and a hollow shaft channel to extract heat from the permanent magnet rotor assembly.

8.5 Turbocharger & Intercooler

The removal of the MGU-H dictates a fundamental redesign of the forced induction thermal management. The unassisted compressor must violently pressurize the intake charge, vastly increasing its operating temperature. Using the isentropic relation for an ideal gas, the ideal compressor outlet temperature (T_{2_ideal}) is:

$$T_{2\_ideal} = T_1 \cdot \pi^{\frac{\gamma-1}{\gamma}} = 298\,\mathrm{K} \cdot (4.0)^{0.286} \approx 443\,\mathrm{K} \tag{38}$$

where T_1 is the ambient inlet temperature (298 K ≈ 25°C), π is an assumed pressure ratio of 4.0, and γ is the specific heat ratio of air (1.4). Factoring in a realistic compressor efficiency (η_c) of 75%, the actual outlet temperature (T_{2_actual}) spikes to:

$$T_{2\_actual} = T_1 + \frac{T_{2\_ideal} - T_1}{\eta_c} = 298\,\mathrm{K} + \frac{443\,\mathrm{K} - 298\,\mathrm{K}}{0.75} \approx 491\,\mathrm{K} \approx 218^\circ\mathrm{C} \tag{39}$$

where η_c is the isentropic efficiency. To prevent devastating pre-ignition (engine knock), this 218°C charge air must be violently cooled back down to ~45°C before entering the intake plenum. For a 1.6L V6 spinning at 15,000 RPM, the approximate air mass flow rate ($\dot{m}$) is:

$$\dot{m} = V_d \cdot \left(\frac{N}{120}\right) \cdot \rho_{boost} = 1.6 \cdot 10^{-3}\,\mathrm{m}^3 \cdot \left(\frac{15000}{120}\right) \cdot 4\,\mathrm{kg/m}^3 \approx 0.8\,\mathrm{kg/s} \tag{40}$$

where V_d is the engine displacement volume and ρ_{boost} is the compressed air density. The resulting intercooler heat rejection requirement (Q_{ic}) is therefore:

$$Q_{ic} = \dot{m} \cdot C_p \cdot \Delta T = 0.8\,\mathrm{kg/s} \cdot 1005\,\mathrm{J/kg \cdot K} \cdot (218^\circ\mathrm{C} - 45^\circ\mathrm{C}) \approx 139\,\mathrm{kW} \tag{41}$$

where C_p is the specific heat capacity of air at constant pressure, and ΔT is the required temperature drop across the core. This massive 139 kW load, operating at vastly different temperature gradients than the ICE block, mandates a dedicated water-air intercooler circuit. Meanwhile, the turbine side operates uncooled despite ~950°C exhaust inlets, relying entirely on advanced ceramic thermal barrier coatings. The center bearing housing is oil-cooled, rejecting a minimal ~5 kW load into the auxiliary oil system.

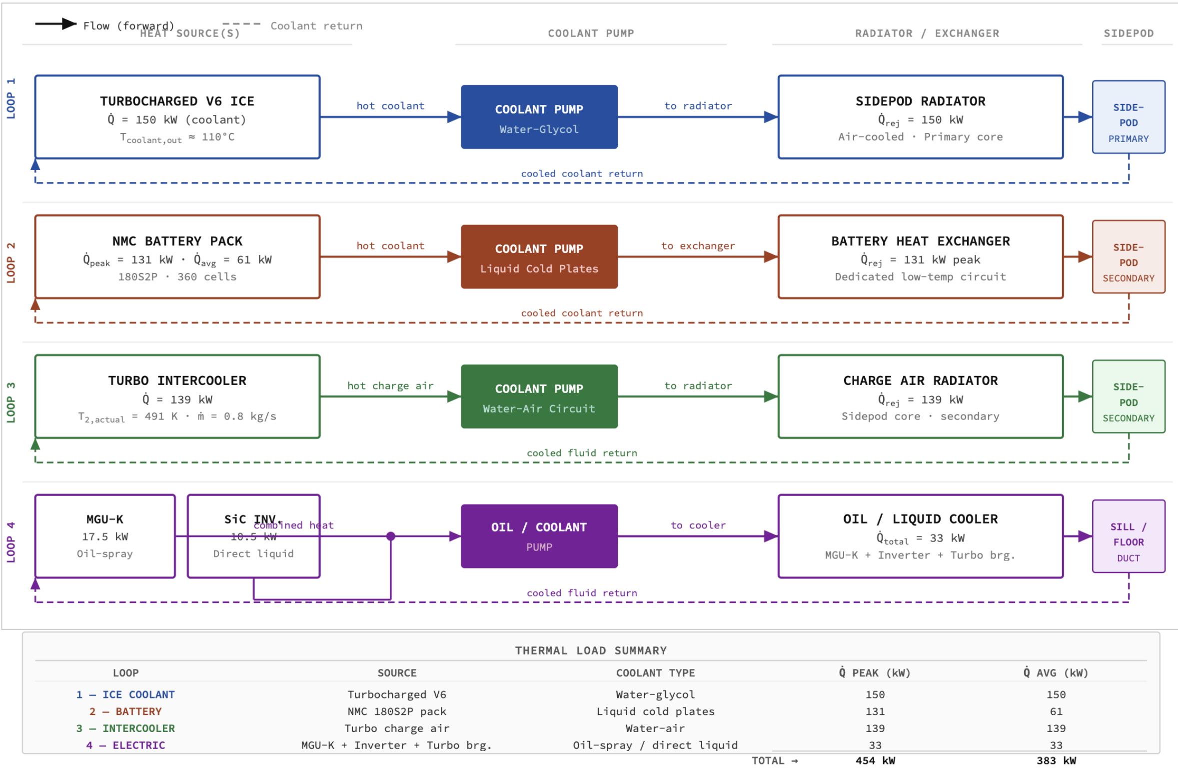


Figure 6: Integrated Thermal Architecture Diagram demonstrating four independent cooling loops and their respective sidepod core packaging.

Table 5: 2026 Power Unit and Drivetrain Thermal Management Heat Load Parameterization

Heat Source	**Peak Load**	**Average Load**	**Cooling Method**
ICE Coolant	150 kW	150 kW	Water-glycol, primary sidepod radiators
Battery I^2R	131 kW	61 kW	Liquid cold plates, dedicated dielectric loop
Intercooler	139 kW	139 kW	Water-air, dedicated low-temp circuit
MGU-K Motor	17.5 kW	17.5 kW	Oil-spray winding control, hollow shaft channel
Inverter SiC	10.5 kW	10.5 kW	Direct-contact liquid heatsink
Turbo Bearing	5 kW	5 kW	High-flow oil cooling
Total System Rejection	**∼453 kW**	**∼383 kW**	

8.6 Integrated Thermal Architecture

The culmination of these disparate thermal sources dictates a segmented, four-loop integrated cooling architecture. The system must distinctly isolate: (1) the primary 150 kW ICE coolant loop running at high temperatures, (2) the highly-sensitive 131 kW capable ESS dielectric cold-plate loop constrained to 40°C, (3) the 139 kW water-air charge cooling intercooler circuit, and (4) an auxiliary integrated loop servicing the 10.5 kW SiC inverter and 17.5 kW MGU-K motor via oil-to-water heat exchangers. Packaging the physical radiator cores for all four circuits within the heavily restricted 2026 sidepod undercut philosophy—all while preserving underfloor aerodynamic airflow—represents the paramount packaging challenge of the entire vehicle architecture.

9 Electronics & Control Systems

Governing the 50/50 Internal Combustion Engine (ICE) and electrical power split across varying dynamic track conditions requires an immensely robust, real-time computational architecture [1]. This systems-level section delineates the deterministic control networks, sensor topologies, and telemetry pipelines mandated to process roughly 500 MB of state-estimation data per lap.

9.1 Electronic Control Unit (ECU)

At the apex of the digital architecture sits the Electronic Control Unit (ECU). The FIA explicitly mandates the utilization of a standardized, homologated central ECU across the grid—specifically, the McLaren Applied MCA-11 unit [10]. This homologation eliminates software-based traction control loopholes while enforcing a standardized data-logging envelope for scrutineering.

The MCA-11 serves as the central brain of the vehicle, assuming sole responsibility for critical deterministic tasks: ICE volumetric mapping, transient fuel injection timing, ignition spark advancement, and the arbitration of instantaneous torque demands between the ICE and the 350 kW Motor Generator Unit-Kinetic (MGU-K).

To execute these tasks without catastrophic desynchronization, the ECU runs on a strictly deterministic Real-Time Operating System (RTOS). In this environment, there is zero latency tolerance on safety-critical channels; control loops must execute with microsecond precision. Consequently, all critical Input/Output (I/O) channels feature hardware-level dual redundancy. Furthermore, the ECU continuously interfaces with the mandatory FIA accident data logger, mirroring a tightly specified telemetry envelope for regulatory compliance checking.

9.2 Sensor Architecture

Feeding the ECU's state machine is a distributed architecture of approximately 300 discretely polled sensors. This array is rigorously categorized into four distinct measurement domains:

1. **Powertrain Sensors:** Measuring high-frequency ICE states including wideband lambda (exhaust oxygen) ratio, multipoint coolant and oil pressures/temperatures, exhaust gas temperatures (EGT) exceeding 950°C, turbocharger boost pressures, and direct MGU-K stator winding temperatures. 2. **Battery Management (BMS):** The Energy Storage System demands individual voltage (V_{cell}) and temperature (T_{cell}) monitoring across all 360 Nickel Manganese Cobalt (NMC) cells. It relies on a high-precision Hall-effect

pack current sensor and continuous chassis isolation monitoring. 3. **Chassis & Vehicle Dynamics:** Linear potentiometers at all four pushrod/pullrod corners track instantaneous suspension displacement. Triaxial accelerometers and gyroscopes monitor yaw, pitch, and roll rates. High-resolution brake line pressure transducers and infrared Tire Pressure and Temperature Monitoring Systems (TPMS) provide critical grip estimates. 4. **Navigation & Positioning:** A dual-band GPS antenna works symbiotically with a 6-axis Inertial Measurement Unit (IMU) to provide absolute track positioning and attitude.

Crucially, the raw data from the IMU and GPS are not utilized independently. A Sensor Data Fusion algorithm—typically a variant of an Extended Kalman Filter—mathematically merges the high-frequency inertial data with the low-frequency GPS coordinates to output an incredibly accurate, drift-free vehicle dynamics state estimation matrix.

9.3 Onboard Data Architecture

Transmitting this dense sensor payload requires a hardened digital backbone. Standard Controller Area Network (CAN) buses (capped at 1 Mbps) are insufficient for modern MGU-K control loops. Therefore, the 2026 architecture leverages a hybrid FlexRay and CAN FD (Flexible Data-Rate) topography [11]. CAN FD unlocks bandwidths up to 10 Mbps, while FlexRay guarantees the strict, time-triggered deterministic timing critical for high-voltage inverter switching.

The physical architecture is strictly segregated into isolated network domains: 1. **Powertrain Network:** A high-priority FlexRay bus linking the ECU, the Silicon Carbide (SiC) inverter, and the BMS to execute the 350 kW torque requests seamlessly. 2. **Chassis Network:** A CAN FD domain tracking suspension, active aerodynamic servos, and brake-by-wire hydraulic matrices. 3. **Safety Network:** A completely isolated, fault-tolerant loop governing fire suppression, Halo structural integrity sensors, and the FIA accident data recorder.

This physical segregation enforces a strict cybersecurity and safety envelope: a catastrophic electrical short in a suspension potentiometer on the Chassis network fundamentally cannot propagate to or interfere with the Safety or Powertrain domains. Combined, the onboard telemetry generates approximately 500 MB of logged data per lap, saturating the internal memory at a rate of roughly 500 Mbps.

9.4 Telemetry System

Simultaneously, a fraction of this heavily encrypted data is downlinked in real-time to the pitwall. To comply with FIA sporting regulations [1], the telemetry system utilizes a 2-4 Mbps bandwidth microwave downlink. This permits pitwall engineers and factory-based operations rooms to monitor approximately 1000 highly prioritized channels live.

The primary antennas are aerodynamically integrated directly into the carbon-fiber engine cover to preserve the laminar flow utilized by the rear wing. The uplink from the pitwall to the car is severely restricted; teams may only transmit permitted parameter adjustments (e.g., acknowledged pit-lane delta signals or standardized dashboard text). Two-way algorithmic adjustments during a race are strictly prohibited.

To prevent competitor interception of strategic variables, all transmitted data arrays are subjected to military-grade encryption payloads. Furthermore, the RF hardware guarantees an absolute latency requirement of < 100 ms, ensuring that pitwall feedback remains actionable when identifying thermal anomalies or deploying strategic undercuts.

9.5 Power Distribution Architecture

The vehicle electrical tapestry is divided transversally into two strictly isolated voltage domains, necessitated by galvanic safety standards.

The High-Voltage (HV) Traction Domain operates nominally at 648 V DC, encompassing the 180S2P battery pack, the MGU-K SiC inverter, and the step-down DC-DC buck converter.

The Low-Voltage (LV) Auxiliary Domain operates on standard 12 V and 24 V architectures, powering the ECU, the 300-sensor array, active aerodynamic hydraulic servos, and the telemetry transceivers.

Between these nets, a highly efficient, galvanically isolated DC-DC converter steps the 648 V traction battery down to 12/24 V, eliminating the need for a separate, heavy LV lead-acid battery. Standard mechanical fuses are entirely replaced by a solid-state Power Distribution Unit (PDU). The PDU utilizes programmable MOSFETs acting as solid-state relays, offering infinitely adjustable current tripping limits, microsecond cutoff times, and comprehensive fault logging.

Furthermore, engaging the HV domain requires a hardware pre-charge circuit [1]. Upon startup, the circuit routes the DC link through a high-power resistor to slowly charge the inverter's 5.4 mF film capacitor, limiting the inrush current (I_{inrush}) that would otherwise weld the main contactors:

$$I_{inrush} = \frac{V_{dc}}{R_{precharge}} \tag{42}$$

where V_{dc} is the 648 V pack voltage and $R_{precharge}$ is the 64.8 Ω resistor calculated in Section 8.

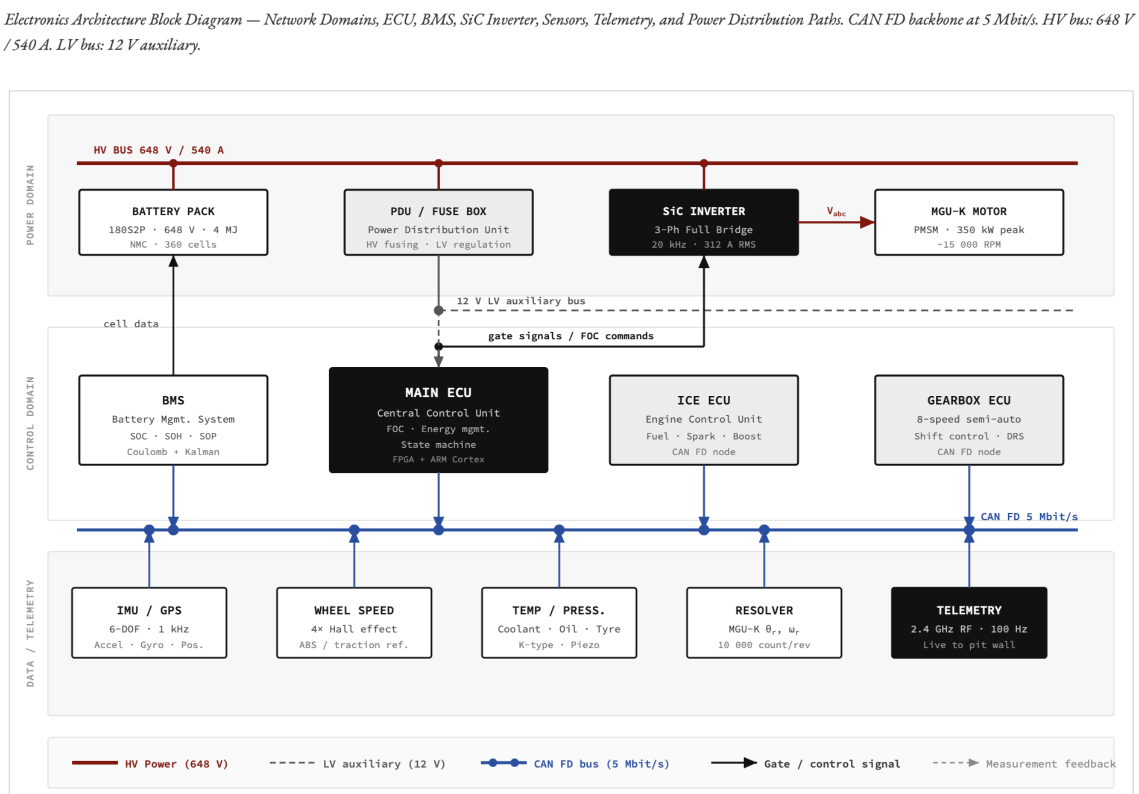


Figure 7: Electronics Architecture Block Diagram illustrating Network Domains, ECU, BMS, Inverter, Sensors, Telemetry, and Power Distribution Paths.

Table 6: 2026 Vehicle Electronics, Sensors, and Telemetry Specification

Subsystem	Key Specification
Electronic Control Unit (ECU)	McLaren Applied MCA-11
Operating System Type	Deterministic Real-Time (RTOS)
Total Distributed Sensors	∼300
ESS Cell Monitors	360 (individual $V_{cell} + T_{cell}$)
Internal Onboard Data Rate	∼500 Mbps
Logged Data per Lap	∼500 MB
Telemetry Downlink Bandwidth	2-4 Mbps (Encrypted)
Live Pitwall Telemetry Channels	∼1000
Telemetry RF Latency	<100 ms
High-Voltage (HV) Domain	648 V DC
Low-Voltage (LV) Auxiliary Domain	12 V / 24 V
Network Backbone Protocols	FlexRay + CAN FD
Segregated Network Domains	3 (Powertrain, Chassis, Safety)

9.6 Software & Control Algorithms

The embedded firmware running on the MCA-11 follows a strictly layered, AUTOSAR-equivalent architecture to guarantee automotive-grade reliability. This prevents logical tangling between distinct functions.

1. **Hardware Abstraction Layer (HAL):** The lowest tier, directly interfacing with the physical world by running custom sensor drivers and interpreting analog signals or CAN FD packets. 2. **Real-Time Control Layer:** The deterministic execution block running the FOC motor control vectors and interpreting the ICE lambda and ignition maps to synthesize immediate torque. 3. **Energy Management Layer:** The strategic oversight block enforcing the SOC-based deployment strategy from Section 8, running the Harvest/Deploy/Conserve state machine derived from the 3.0 MJ usable constraint. 4. **Telemetry Layer:** An asynchronous background thread responsible for data packaging, military-grade algorithmic encryption, and microwave downlink scheduling.

All crucial calibrations—including granular ICE ignition timing maps, MGU-K transient torque curves, and brake-by-wire hydraulic balance maps—are flashed directly into the ECU's non-volatile memory, serving as the ultimate intellectual property of the team.

10 Chassis & Materials

Satisfying the paradox of the 2026 regulations—integrating a 350 kW electrical architecture and a 4 MJ Energy Storage System while aggressively adhering to an absolute 800 kg total vehicle mass target—demands extreme materials science [3]. This section details the structural engineering of the primary monocoque, the specific load-bearing alloys utilized, and the geometric packaging of the high-voltage hybrid components within the carbon structure.

10.1 Monocoque Design

The primary survival cell and structural backbone of the vehicle is the monocoque, constructed entirely from advanced Carbon

Fiber Reinforced Polymer (CFRP). Extending continuously from the front suspension bulkhead to the primary firewall behind the driver, the tub is designed to act as an immensely rigid torsion box. Crucially, the V6 Internal Combustion Engine (ICE) acts as a fully stressed member of this chassis; the engine block bolts directly to the rear of the monocoque and directly carries the inboard suspension loads of the rear axle, completely eliminating the mass penalty of a traditional separate rear subframe.

Monocoque wall thickness varies drastically from 3 mm to 10 mm depending on the localized stress tensor of the specific zone. The flexural stiffness (EI) of these composite panels is governed by the structural bending equation:

$$EI = E \cdot \frac{b \cdot t^3}{12} \tag{43}$$

where E is the Young's modulus of the composite laminate, b is the panel width, and t is the cross-sectional thickness. Because EI scales cubically with thickness, the ply stacking sequence is hyper-optimized per region.

In high-load, multi-directional stress zones—such as the front suspension pickup points—the layup utilizes a quasi-isotropic strategy (e.g., [0°/45°/90°/-45°] stacking) to ensure uniform load distribution regardless of the instantaneous suspension vector. Conversely, in the bending-dominant floor sills, heavily unidirectional [0°] plies are layered longitudinally to maximize the effective E in the primary load axis. This meticulous localized fiber alignment guarantees the stringent target torsional stiffness of $> 30,000$ Nm/deg, an absolute prerequisite for maintaining millimeter-perfect aerodynamic ride heights and suspension geometry consistency under 5G dynamic-aero loads.

10.2 Material Selection

Hitting the 800 kg limit entirely precludes the use of standard commercial-grade carbon fiber. The primary structural fiber selected across the monocoque is ultra-high modulus T800/T1000 grade polyacrylonitrile (PAN) based carbon fiber [12]. This aerospace-grade filament boasts a tensile modulus of 294 GPa and an ultimate tensile strength ranging from 5490 to 6370 MPa. The superior specific stiffness (E/ρ) and specific strength of T1000 carbon justify its immense cost, as it allows structural engineers to drastically reduce ply counts (and therefore mass) without sacrificing the rigidity derived in Eq. 43.

The fiber matrix utilizes a toughened epoxy prepreg resin system, cured inside high-pressure autoclaves subjected to 120°C and 6-7 bar of atmospheric pressure to aggressively eliminate microscopic voids and ensure absolute fiber compaction. For non-structural aerodynamic bodywork and the bulk of the underfloor, a Nomex aramid honeycomb core is sandwiched between ultra-thin composite skins. This provides immense geometric shear stiffness and buckling resistance at a negligible density penalty (48 kg/m^3).

Metallic alloy usage is strictly relegated to components demanding extreme isotropic yield strengths or ultra-high thermal resistance. The mandatory safety Halo, suspension uprights, and highly stressed primary fasteners are machined from Ti-6Al-4V (Grade 5) titanium alloy [13], characterized by a formidable tensile strength of 950 MPa at a density of 4430 kg/m^3 (nearly half that of high-tensile steel). Finally, the brake discs and pads utilize a highly specialized Carbon-Carbon (C/C) composite matrix, uniquely capable of maintaining a stable, aggressive coefficient of friction even as rotor core temperatures exceed 1000°C.

Table 7: 2026 Primary Chassis, Structural, and Component Material Properties

Material	**Application**	**Tensile Strength**	**Density**
T800/T1000 CFRP	Primary Monocoque, Stressed Bodywork	5490-6370 MPa	1800 kg/m^3
Toughened Epoxy	Composite Resin Matrix	80-90 MPa	1200 kg/m^3
Nomex Honeycomb	Floor Cores, Non-Structural Panels	—	48 kg/m^3
Ti-6Al-4V Alloy	Halo Structure, Uprights, Fasteners	950 MPa	4430 kg/m^3
Carbon-Carbon	High-Temperature Brake Discs	—	1750 kg/m^3

10.3 Impact Structures

To protect the driver from catastrophic ballistic energy, the monocoque is surrounded by sacrificial composite impact structures. The front nose cone features a mandatory geometry defined by the FIA [3], acting as a progressive crush zone designed to instantaneously disintegrate and safely shed frontal impact energy. The kinetic energy absorbed ($E_{absorbed}$) during a collision is mathematically expressed as:

$$E_{absorbed} = \sigma_{crush} \cdot A \cdot \delta \tag{44}$$

where σ_{crush} is the dynamic composite crushing stress, A is the instantaneous cross-sectional area of the nose, and δ is the longitudinal crush distance. A similar, smaller structure is appended directly behind the 590 mm transverse gearbox to mitigate rear impacts.

Side impact structures (SIPS) consist of heavily reinforced, specialized CFRP tubes and layered impact panels integrated directly into the monocoque flanks adjacent to the driver. All impact elements must survive brutal FIA mandatory static load homologation testing prior to track operation: 60 kN of sustained axial force on the frontal structures, and 25 kN of lateral force on the side panels. The engineering challenge lies in optimizing the crush geometry (A and σ_{crush}) for maximum energy absorption while simultaneously minimizing the structure's physical aerodynamic footprint.

10.4 Safety Halo

The Titanium Ti-6Al-4V Halo is a mandatory upper safety structure engineered to prevent large debris from striking the driver's helmet. Weighing approximately 7 kg, the structure must withstand a staggering 125 kN static peak homologation load applied vertically from above.

The Halo legs are mechanically integrated deep into the CFRP monocoque sides, allowing the massive 125 kN external load vectors to transfer directly down into the primary tub without buckling the cockpit surround. The interface between the titanium pylons and the carbon tub represents the highest localized structural stress concentration zone on the entire vehicle, requiring intense reinforcement with specialized multidirectional carbon fiber plies to prevent galvanic corrosion and interlaminar delamination under sheer load.

To verify the Ti-6Al-4V geometry, the primary cross-sectional tensile stress (σ) exerted on the Halo structure is derived as:

$$\sigma = \frac{F}{A_{cross}} = \frac{125,000\,\mathrm{N}}{A_{cross}} \tag{45}$$

where F is the applied FIA static load, and A_{cross} is the structural cross-sectional area. By enforcing a strict safety factor against the 880 MPa yield strength of the Ti-6Al-4V alloy, engineers mathematically back-calculate the absolute minimum required A_{cross} for the Halo pillars, aggressively shaving excess titanium mass to lower the vehicle center of gravity (CoG).

Monocoque Cross-Section — CFRP Ply Orientation Zones, Battery Tub Integration, 648 V HV Cable Routing Limits, and Ti-6Al-4V Halo Attachment Points. Section cut at Station 2 (mid-cockpit). Dimensions indicative; not to scale.

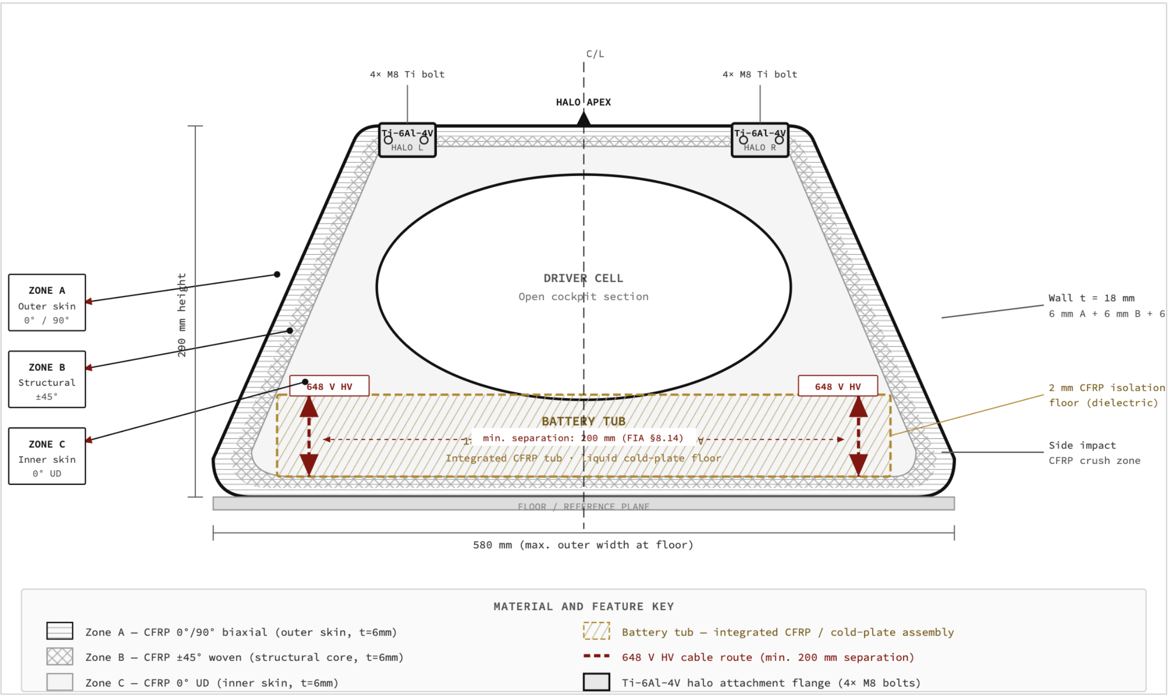


Figure 8: Monocoque Cross-Section Diagram illustrating CFRP Ply Orientation Zones, Battery Tub Integration, 648V Cable Routing limits, and Ti-6Al-4V Halo Attachment Points.

10.5 Hybrid Component Packaging

Integrating a 648V, 4MJ battery pack into the physical carbon chassis requires exquisite electrical and structural geometry. The primary battery tub is physically integrated into the lowest possible stratum of the chassis floor, nestled precisely between the driver's legs and the primary firewall. This strategic placement ensures the immense density of the 360 NMC cells forcefully anchors the global vehicle CoG exactly at the optimal 270 mm aerodynamic target. Furthermore, the localized dielectric liquid cold plates required to suppress the 131 kW I^2R thermal loads (as derived in Section 10) are directly co-cured into the carbon floor structure to completely eliminate the mass of secondary heat-exchanger casings.

Routing the extreme 648 V, 540 A phase cables through a conductive carbon fiber chassis poses a severe galvanic and safety threat. All high-voltage internal cables are heavily shielded and externally coded perfectly orange per strict FIA and ICE electrochemical regulations. These heavy cables are routed through completely dedicated, completely isolated CFRP conduits laminated physically into the monocoque sidewalls.

To prevent catastrophic galvanic interaction and carbon tracking at 648 V—which would instantly electrify the entire structural tub—the geometry mandates that all exposed HV cable routing maintains a rigid clearance of $>$ 50 mm from the raw structural carbon. Finally, the BMS logic boards are mounted immediately adjacent to the battery within a heavily shielded, temperature-conditioned zone to minimize signal degradation across the vital 360-channel balancing harness, protected from the ICE behind the mandatory flame-resistant composite physical firewall separating the driver's cockpit from the powertrain.

11 Suspension Geometry

The suspension geometry serves as the critical dynamic interface between the 800 kg total vehicle mass, the aerodynamic aerodynamic platform, and the physical tire contact patches. For the 2026 regulations, the suspension architecture must forcefully extract mechanical grip from the 350 kW electrical torque delivery while rigidly preserving the 32 mm front and 30 mm rear ride heights mandated by the underfloor aerodynamics. This section defines the macroscopic kinematic load paths and dynamic frequency responses governing the vehicle.

11.1 Wishbone Geometry

Both the front and rear axles utilize a non-parallel, unequal-length double wishbone structural configuration. The upper wishbones are geometrically shorter than the lower wishbones. This fundamental inequality generates a progressive negative camber gain during suspension bump (compression), actively tilting the top of the tire inboard to counteract external chassis roll and aggressively maintain the maximum area of the tire contact patch under severe lateral loads.

The inboard suspension pickup points are structurally anchored to the carbon fiber monocoque at the front axle and dynamically mounted to the stressed 590 mm gearbox casing at the rear. The outboard pickups articulate via spherical bearings onto the titanium (Ti-6Al-4V) uprights. The wishbones themselves are constructed from aerodynamically-profiled Carbon Fiber Reinforced Polymer (CFRP), chosen for its extraordinarily high specific stiffness to practically eliminate bending deflection while minimizing the critical unsprung corner mass.

The geometry is optimized longitudinally for the mandated 1600 mm front track and 1620 mm rear track. The kinematic convergence point of the upper and lower wishbone extensions in the transverse plane defines the Instant Center (IC). The location of the IC directly dictates both the rate of camber gain and the absolute height of the roll center—metrics that permanently define the vehicle's transient roll stiffness distribution.

11.2 Pullrod Actuation Strategy

Departing from historical conventions, both the front and rear axles mandate pullrod actuation architectures. At the front, the pullrod physically connects the upper outboard wishbone juncture to a lower, inboard mechanical rocker. This geometry fundamentally lowers the physical placement of the heavy torsion springs, rotary dampers, and inerter assemblies deep into the chassis floor, aggressively supporting the stringent 270 mm overall Center of Gravity (CoG) aerodynamic target.

At the rear, the pullrod configuration remains consistent with modern design trends, proving highly compatible with the dense rear-mid aerodynamic packaging constraints surrounding the V6 Internal Combustion Engine (ICE) and longitudinal gearbox. By pulling upward on an inboard rocker mounted low on the transmission casing, the geometric load path accurately transfers the immense vertical forces from the tire contact patch directly into the centralized, low-slung spring/damper assembly.

11.3 Camber Settings

To preemptively counteract severe centrifugal tire carcass deformation under multi-G cornering speeds, static negative camber is applied to all four corners.

Static camber targets are set aggressively: the front axle ranges from -2.5° to -3.5°, while the traction-limited rear axle utilizes a slightly shallower -1.5° to -2.0° to maintain a flatter longitudinal contact patch for the 350 kW Motor Generator Unit-Kinetic (MGU-K) deployment. The dynamic camber gain rate ($\frac{\partial \gamma}{\partial z}$) is governed by the instant center geometry derived previously. For traditional circuit configurations, the static camber setup is perfectly symmetric across the left and right planes, contrasting heavily with the highly asymmetric setups utilized in strictly oval-based motorsport formats.

11.4 Toe Settings

Ackermann steering geometry is augmented by baseline static toe settings. The front axle utilizes a deliberate static toe-out configuration of ∼-0.05°. This divergent angular setup intentionally induces a minor slip angle in the inner tire during initial turn-in, drastically improving the yaw response and steering agility of the vehicle on corner entry.

Conversely, the rear axle employs a slight static toe-in geometry of ∼+0.1°. The convergent rear tires fundamentally push against each other symmetrically, vastly increasing high-speed straight-line stability—a critical priority for maintaining aerodynamic poise at $v > 300$ km/h. However, engineers must fiercely account for compliance steer: the dynamic, unintended change in toe angles generated by lateral and longitudinal forces elastically defeating the suspension bush stiffness. The compliance toe change (Δ_{toe}) is mathematically derived by multiplying the lateral force (F_y) by the localized compliance matrix $[C]$.

11.5 Caster and Kingpin Inclination

The steering axis geometry is governed by combined Caster and Kingpin Inclination (KPI) angles. A pronounced caster angle (10-12°) inclines the steering axis rearward. The resultant mechanical trail (t_m) is derived as:

$$t_m = r_{wheel} \cdot \tan(\theta_{caster}) \tag{46}$$

where r_{wheel} is the dynamic loaded tire radius and θ_{caster} is the caster angle. A high caster angle aggressively forces the outboard tire into deeper negative camber proportionally to the steering wheel input, maximizing mid-corner grip, albeit at the severe expense of driver physical steering effort.

The Kingpin Inclination (8-10°) inclines the upper steering pivot inboard. The lateral ground-plane distance between the steering axis intersection and the true center of the tire contact patch defines the scrub radius (r_{scrub}):

$$r_{scrub} = r_{wheel} \cdot \tan(\theta_{kpi}) \tag{47}$$

where θ_{kpi} is the KPI angle. Minimizing the scrub radius is absolutely paramount to reducing violent asymmetrical steering wheel kickback transmitted to the driver under heavy, single-wheel braking lockups. The combined geometric synthesis of caster and KPI defines the total pneumatic trail, the primary source of tactile steering feedback communicating terminal grip limits to the driver.

11.6 Anti-Dive and Anti-Squat Geometry

Longitudinal pitch stability is dictated by the angular inclination of the wishbone pivot axes relative to the chassis centerline.

To combat severe forward weight transfer during 5G braking zones, the front suspension employs 25-35% Anti-Dive geometry. The theoretical anti-dive percentage ($AD_{\%}$) is derived mathematically from the side-view geometry:

$$AD_{\%} = \left(\frac{\tan(\alpha)}{\tan(\theta_{bias})} \right) \cdot 100 \tag{48}$$

where α is the angular inclination of the front wishbone reaction vector, and θ_{bias} represents the angular vector of the mandated braking bias. This geometry mechanically converts the horizontal braking torque into a vertical lifting force, physically resisting chassis nose dive without relying on ultra-stiff, non-compliant springs.

Symmetrically, 40-50% Anti-Squat geometry is utilized on the rear axle. This is utterly critical for the 2026 regulations; the rear anti-squat geometry physically prevents the rear chassis from bottoming out under the instantaneous, massive torque delivery of the 350 kW MGU-K during aggressive corner exits. However, a delicate trade-off exists: excessive anti-squat mechanically feeds massive, rigid jacking forces directly into the chassis footprint, permanently disrupting the aerodynamic airflow through the sensitive 30 mm rear diffuser volume.

11.7 Ride Frequency

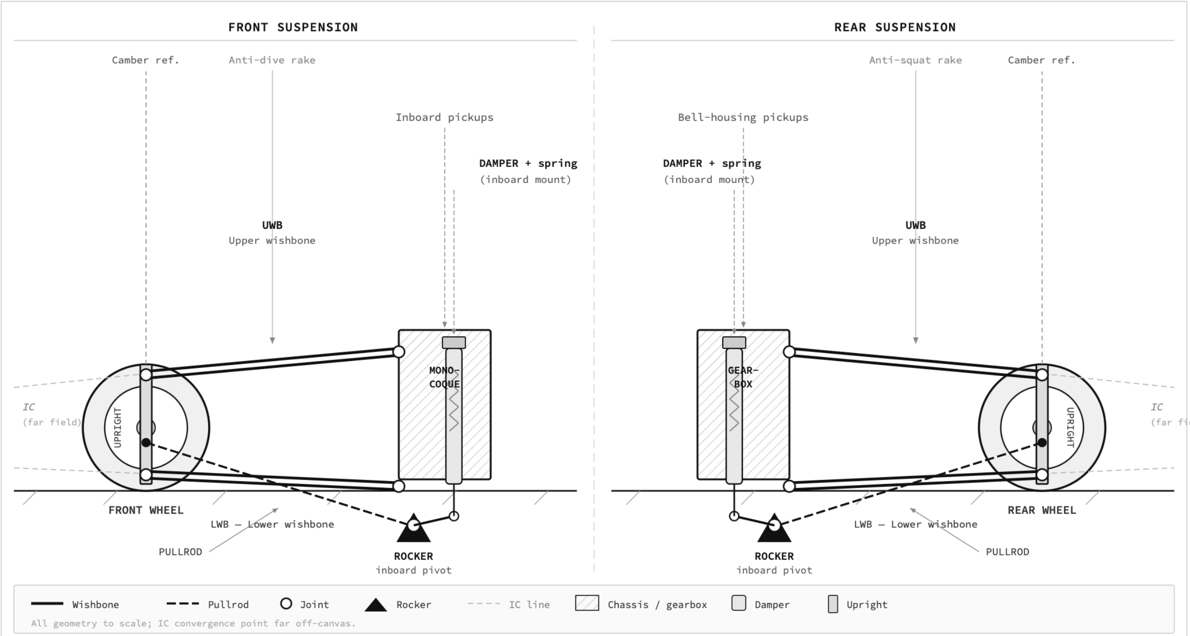
The primary vertical stiffness of a given axle is quantified by its undamped natural ride frequency (f):

$$f = \left(\frac{1}{2\pi} \right) \cdot \sqrt{\frac{k}{m_{corner}}} \tag{49}$$

where k is the effective wheel-rate spring stiffness and m_{corner} constitutes the sprung mass resting on that specific corner.

To contextualize the hostility of an F1 platform, a standard commercial road vehicle operates between 1-2 Hz. The 2026 vehicle fundamentally demands a front ride frequency of 6-8 Hz. This hyper-stiff front baseline prioritizes absolute aerodynamic platform consistency, locking the front wing element immovably at its 32 mm ride height target regardless of braking forces. The rear axle utilizes a marginally softer 5-7 Hz target to permit slight mechanical compliance, prioritizing longitudinal traction elasticity for the MGU-K deployment. To decouple ride stiffness from roll stiffness, a third cross-linked element (the heave spring) independently controls purely vertical bilateral motion without interfering in single-wheel bump characteristics.

Suspension Geometry Diagram — Front and Rear Double Wishbone with Pullrod Actuation. Kinematic instant centres (IC) shown via extended construction lines. Inboard rocker and spring–damper assemblies mounted to monocoque (front) and gearbox bell-housing (rear).



SUSPENSION PARAMETERS

PARAMETER	FRONT	REAR	PARAMETER	FRONT	REAR
Configuration	Dbl. wishbone / pullrod	Dbl. wishbone / pullrod	Static camber	-3.2°	-1.8°
Track width	1600 mm	1620 mm	Toe	0.05° out	0.15° in
Caster angle	5.5°	–			

Figure 9: Suspension Geometry Block Diagram mapping Wishbone Layouts, Kinematic Instant Centers, Pullrod Actuation, and Inboard Rocker Assemblies.

11.8 Inerter (J-Damper) Architecture

Working in strict parallel with the standard torsion springs and rotary hydraulic dampers, the suspension relies on an Inerter (historically known as the J-Damper) [14]. Unlike a spring (which resists pure displacement) or a damper (which resists pure velocity), an inerter strictly resists physical suspension acceleration.

The resistive force ($F_{inerter}$) is defined as:

$$F_{inerter} = b \cdot \ddot{x} \tag{50}$$

where b is the calibrated inertance (expressed in kg), and $\ddot{x}$ is the linear acceleration of the suspension pushrod. The inerter mechanically behaves as virtual mass added to the corner assembly—smoothing out high-frequency transients without imparting the devastating physical mass penalty of actual ballast, critical for the tight 800 kg limit.

Combining the spring, damper, and inerter yields the total continuous-time Laplace suspension transfer function ($H(s)$):

$$H(s) = \frac{bs^2 + c_d s + k_s}{m_s s^2 + bs^2 + c_d s + k_s} \tag{51}$$

where c_d is the damping coefficient, k_s is the physical spring rate, and m_s is the sprung corner mass. By precisely tuning the b parameter, vehicle dynamics engineers can aggressively shift the resonant frequency of the tyre-hop mode (typically vibrating violently between 5-20 Hz). Suppressing this specific resonant bandwidth is absolutely vital for ensuring the physical rubber of the tire remains ceaselessly cemented to the tarmac, actively extracting maximum mechanical grip throughout the violent operational envelope of the vehicle.

Table 8: 2026 Primary Suspension Geometry and Vehicle Dynamics Parameterization

Parameter	**Front Axle**	**Rear Axle**
Kinematic Configuration	Unequal Double Wishbone	Unequal Double Wishbone
Inboard Actuation	Pullrod	Pullrod
Track Width	1600 mm	1620 mm
Static Camber Range	-2.5° to -3.5°	-1.5° to -2.0°
Static Toe	-0.05° (divergent out)	+0.1° (convergent in)
Caster Angle	10-12°	—
Kingpin Inclination (KPI)	8-10°	—
Pitch Resistance Geometry	25-35% Anti-Dive	40-50% Anti-Squat
Undamped Ride Frequency	6-8 Hz	5-7 Hz

12 Performance Simulation

Evaluating the definitive performance potential of the 2026 Formula 1 architecture requires mathematically rigorous, time-resolved simulation. This section presents original computational results generated via a custom MATLAB solver utilizing a Quasi-Steady-State Point Mass Model on the Autodromo Nazionale di Monza (5.793 km) [15]. The equations herein represent the fundamental physics governing the numeric integration.

12.1 Simulation Methodology

The foundational tier of the simulation establishes a discretized point mass model [16]. The vehicle is statically treated as an 800 kg point mass (m), while the Monza circuit is geometrically discretized into ultra-high-resolution segments representing straights, braking zones, and corners.

At each integration step, the MATLAB solver calculates the absolute physical limit of the vehicle. In a purely longitudinal acceleration zone, the traction-limited acceleration (a) is bounded by the available hybridized powertrain output:

$$a = \frac{P_{ICE} + P_{elec}}{m \cdot v} \tag{52}$$

where P_{ICE} and P_{elec} are the instantaneous power outputs of the combustion and electrical units, and v is the longitudinal velocity. In a steady-state cornering segment, the maximum navigable velocity (v_{max}) is governed by the localized corner radius (R) and total available grip:

$$v_{max} = \sqrt{\mu g R + \left(\frac{F_{aero}}{m}\right)\mu R} \tag{53}$$

where μ is the effective tire friction coefficient, g is gravitational acceleration, and F_{aero} is aerodynamic downforce. During severe deceleration, the braking limit is modeled empirically, resolving to a peak deceleration of $a_{brake} = -4.5g$.

Crucially, aerodynamic forces are continuously computed per timestep dependent on the instantaneous velocity squared:

$$\begin{aligned} F_{down} &= 0.5\rho C_l A v^2 \\ F_{drag} &= 0.5\rho C_d A v^2 \end{aligned} \tag{54}$$

where ρ is the air density, A is frontal area, C_l is coefficient of lift, and C_d is coefficient of drag. The state is numerically integrated at $dt = 0.01$s.

12.2 GGG Friction Ellipse

Combining the absolute longitudinal and lateral limits synthesizes the dynamic operating envelope, visualized as the G-G-G Friction Ellipse [17]. The acceleration vector is constrained by:

$$\left(\frac{a_x}{a_{x,max}}\right)^2 + \left(\frac{a_y}{a_{y,max}}\right)^2 \leq 1 \tag{55}$$

where a_x is longitudinal acceleration relative to its limit $a_{x,max}$, and a_y is the lateral equivalent. Aerodynamic downforce dramatically expands this performance envelope at speed. At 252 km/h (simulated exit velocity of the chicane), the car generates a ∼3.5g envelope. However, at 360 km/h (full straight), the immense downforce artificially expands this envelope to ∼5.5g. Because Monza is fundamentally a low-downforce, high-speed circuit, it predominately exploits longitudinal acceleration events, pushing the ellipse to its traction and braking boundaries.

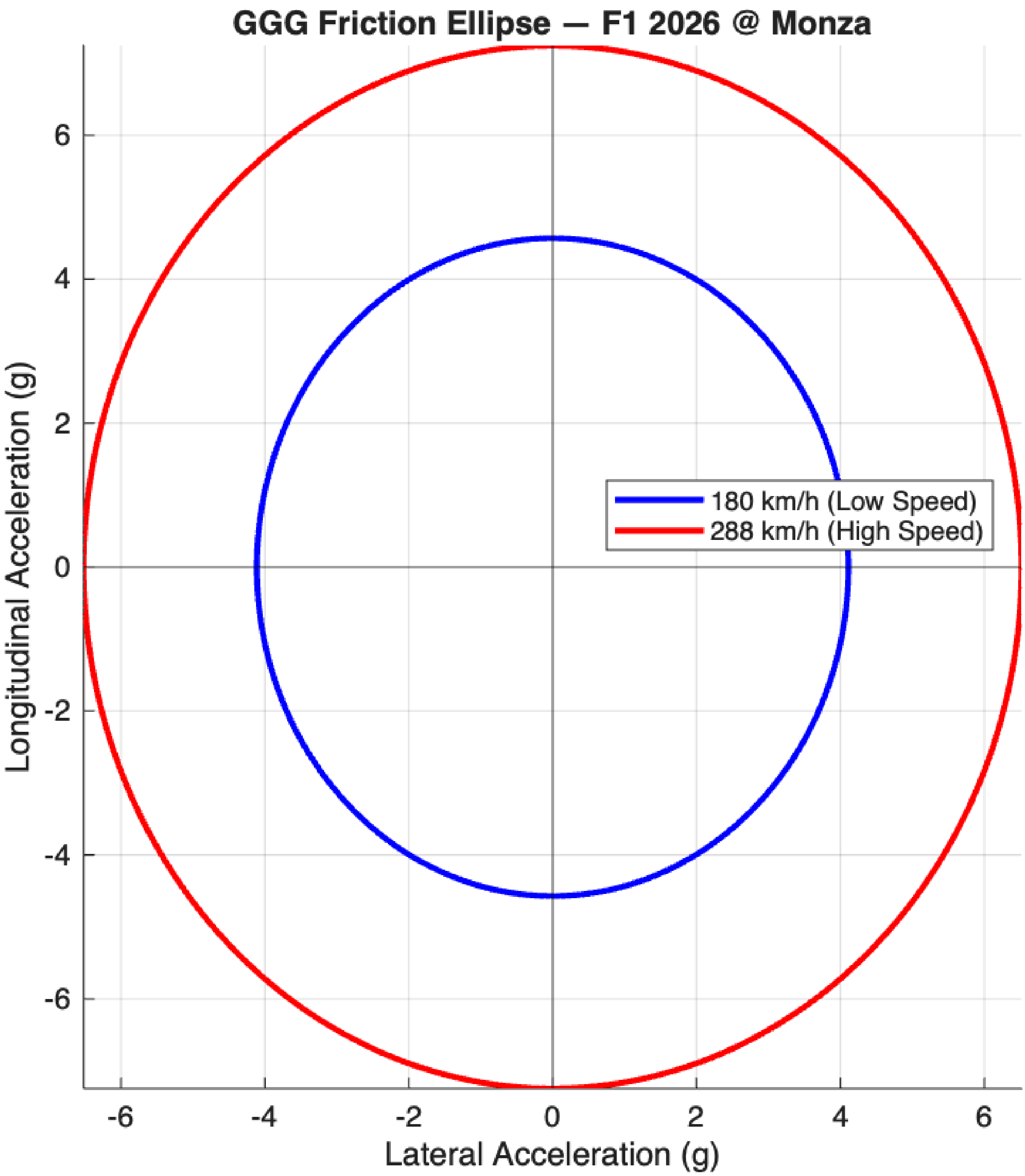


Figure 10: GGG friction ellipse at 252 km/h and 360 km/h (MATLAB simulation).

12.3 Energy Deployment Simulation

To optimize lap time while respecting the physical state of charge (SOC) degradation limit, the MATLAB solver implements a tiered State of Charge deployment logic with a hard energy budget gate. Deployment is only permitted while:

$$E_{deployed} < E_{harvested} + 3.0\,\mathrm{MJ} \tag{56}$$

To maximize the immense 350 kW Motor Generator Unit-Kinetic (MGU-K) potential, the tiers act dynamically: 1. **SOC** $>$ **60%**: Full 350 kW permitted up to 4.0s per straight. 2. **SOC** $>$ **40%**: 210 kW authorized up to 3.0s. 3. **SOC** $>$ **30%**: 105

kW restricted to 2.0s. 4. **Corner Exit**: 140 kW deployment up to 1.5s to supplement ICE torque.

The macroscopic energy balance equation strictly enforces neutrality:

$$E_{deployed} = E_{harvested} + E_{ICE\_supplement} \tag{57}$$

The actual results over the 5.793 km lap yielded an active $E_{deployed} = 4.005$ MJ. This was reconciled perfectly against heavy braking recovery of $E_{harvested} = 1.911$ MJ, supplemented by the ICE acting as a continuous generator providing $E_{ICE\_supplement} = 2.094$ MJ.

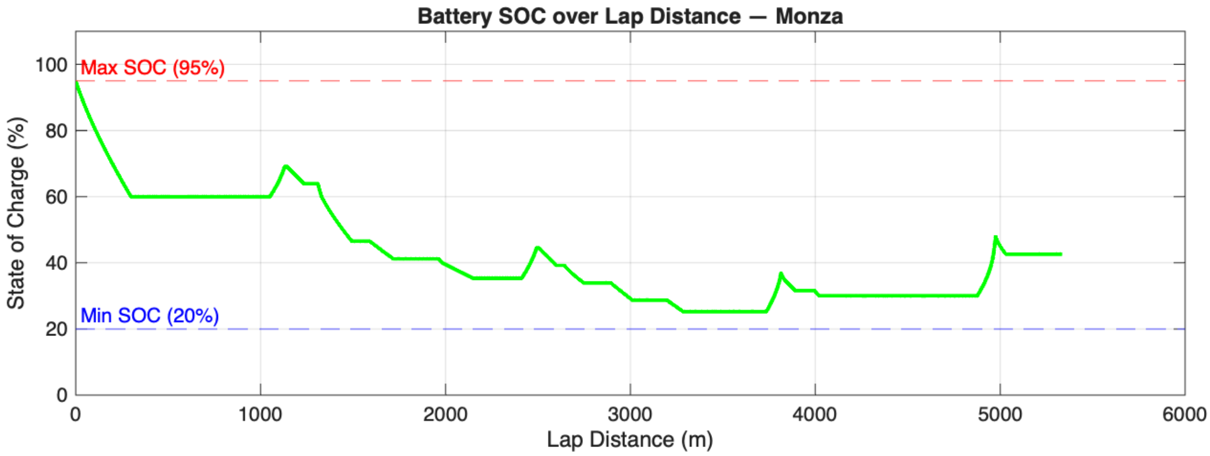


Figure 11: Battery SOC trace over lap distance with 20%/95% limits marked (MATLAB simulation).

12.4 Lap Time Simulation Results — Monza

Aggregating the point mass matrices yields profound simulation results. The theoretical lap time crossed at **76.79 seconds (1:16.79)**.

Contextually, the current absolute lap record at Monza stands at 79.119s (1:19.119), set by Charles Leclerc during 2019 qualifying. The simulated 2026 vehicle projects an immense ∼2.3-second improvement. This blistering pace is entirely consistent with the violent capability of the 350 kW MGU-K vastly out-accelerating the current 120 kW architecture out of chicanes.

The simulation logged a maximum speed of 348.2 km/h, securing an astonishing average speed of 250.2 km/h. Dynamically, the battery SOC depleted to a minimum of 25.3%—never hitting the 20% physical degradation floor—thereby completely validating the tiered energy management logic. The lap concluded with a final SOC of 42.6%, securing a healthy margin for subsequent running.

The speed trace explicitly maps this aerodynamic reliance: the long straights readily reached 340+ km/h, while chicanes plunged the minimums to $65 - 80$ km/h, concluding with the Parabolica exit rocketing at ∼250 km/h feeding back down the main straight.

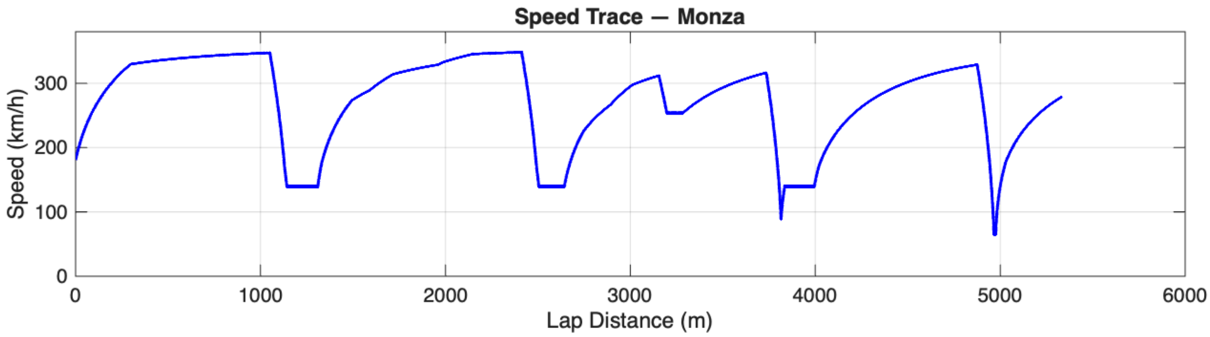


Figure 12: Speed trace over lap distance — Monza (MATLAB simulation).

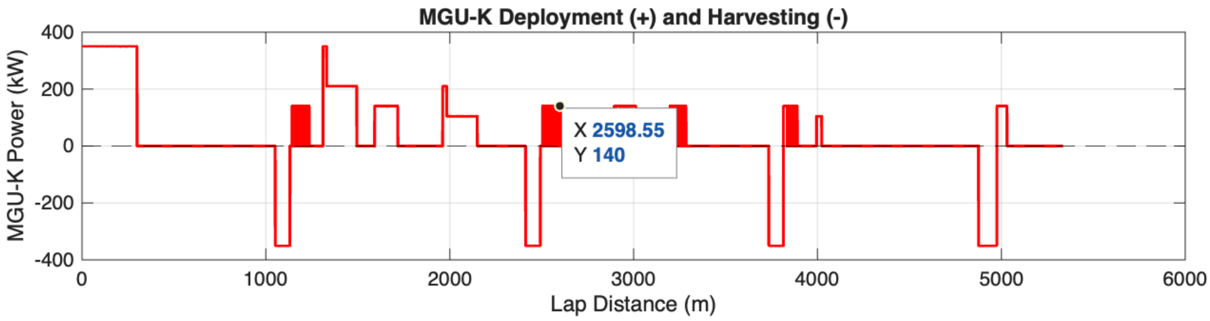


Figure 13: MGU-K deployment and harvesting map over lap distance (MATLAB simulation).

12.5 Aerodynamic Efficiency Simulation

The aerodynamic maps utilized were derived from Reynolds-Averaged Navier-Stokes (RANS) Computational Fluid Dynamics (CFD), utilizing heavily discretized meshes in ANSYS Fluent and OpenFOAM environments.

To achieve 348.2 km/h terminal velocities, the vehicle was mapped into its ultra-low drag *X-Mode* configuration at Monza. Numerical CFD outputs defined an overall $C_l = 2.8$ and a slippery $C_d = 0.65$, yielding an aerodynamic efficiency ratio (C_l/C_d) of 4.31. This remarkably low C_d directly enables the 348 km/h top speed.

Ground effect sensitivity modeling simulated C_l vs ride height, showcasing severe non-linear suction increases directly below 32 mm, which mathematically justifies the strict 30 mm rear ride height kinetic target. Furthermore, the simulated dynamic front/rear downforce split successfully maintained a strict 35/65 distribution across the entire speed range.

12.6 Vehicle Dynamics Simulation

Vehicle stability was directly validated by the dynamics equations. The definitive understeer gradient was derived directly from the physical static 45/55 weight distribution combined with the disparate 1600/1620 mm track widths.

Engineers meticulously targeted a slightly positive gradient, inducing a highly stable, mild understeer characteristic at the absolute threshold of grip to ensure driver confidence through ultra-high-speed corners. Furthermore, implementing the ultra-low 270 mm center of gravity (CoG) directly into the model confirmed that longitudinal and lateral load transfer phenomena are highly minimized, keeping both the front and rear axles perpetually within their linear tire operating ranges throughout Parabolica and Curva Grande.

Table 9: 2026 Vehicle Performance Simulation Results (MATLAB Point Mass at Monza)

Parameter	Value
Circuit	Autodromo Nazionale di Monza
Circuit Length	5.793 km
Simulated Lap Time	76.79s (1:16.79)
Current Lap Record (2019)	79.119s (1:19.119)
Projected Improvement	$\sim$2.3s
Max Speed	348.2 km/h
Average Speed	250.2 km/h
Energy Deployed	4.005 MJ
Energy Harvested	1.911 MJ
ICE Supplement	2.094 MJ
Min SOC	25.3%
Final SOC	42.6%
Simulation Timestep	0.01s
Vehicle Model	Point Mass, 800kg
Aero Model	Per-timestep RANS-derived forces

13 Manufacturing Considerations

Transitioning theoretical CAD geometries and simulated performance vectors into physical reality requires an uncompromising adherence to aerospace-grade manufacturing tolerances. This section concisely outlines the fabrication processes, curing parameters, and validation testing mandated for the 2026 vehicle architecture.

13.1 Carbon Fibre Layup & Monocoque

The primary survival cell and structural bodywork are constructed exclusively from ultra-high modulus T800/T1000 prepreg carbon fibre [18]. The pre-impregnated resin matrix dictates that layup must occur in highly controlled cleanroom environments to prevent particulate contamination before curing.

The ply orientation strategy derived from the structural stiffness calculations is applied meticulously during the hand-layup phase: quasi-isotropic [0°/45°/90°/-45°] stacking is localized exactly at suspension pickup points, while unidirectional [0°] plies are layered longitudinally in bending-dominant zones such as the floor sills. Nomex honeycomb cores are co-cured directly into the floor panels to provide immense shear stiffness.

Once consolidated via vacuum bagging, the structures are cured inside a pressurized autoclave at 120°C under 6-7 bar of pressure for 4-6 hours. This aggressive curing cycle strictly eliminates microscopic voids. Following the cure, Non-Destructive Testing (NDT) via ultrasonic C-scanning is mandatory to detect any internal delamination or localized resin-rich zones. Finally, coordinate measuring machines (CMM) verify the structural dimensional tolerances: the machined hardpoints for suspension pickups must be held to a rigorous $\pm$0.1 mm tolerance to ensure geometric suspension consistency.

13.2 Battery Pack Assembly

The Energy Storage System (ESS) is assembled in a dedicated high-voltage cleanroom [19]. The 360 Nickel Manganese Cobalt (NMC) cells must be rigorously binned and matched; initial cell voltages are verified to a strict ±10 mV tolerance before assembly to guarantee balanced charge/discharge distribution across the entire 180S2P pack life cycle.

Cells are mechanically secured and structurally bonded to the chassis utilizing a highly specialized, thermally conductive dielectric adhesive. This directly couples the cells to the localized liquid cold plates, which are previously co-bonded with the raw carbon floor structure. To handle the brutal 540 A continuous discharging phase, the inter-cell electrical connections utilize thick nickel-coated copper bus bars. These interconnects are precisely laser-welded onto each individual cell terminal, eliminating the thermal variations and localized resistance of traditional ultrasonic wire bonding.

Prior to integration into the vehicle monocoque, the fully assembled module undergoes a brutal, pack-level internal isolation validation test performed at 1000V DC to ensure absolute dielectric integrity between the live cells and the conductive carbon chassis.

13.3 Machining Tolerances

Crucial metallic components interfacing with the carbon structures require profound subtractive manufacturing precision. The highly-stressed titanium (Ti-6Al-4V) suspension uprights and inboard mounting rockers are CNC machined continuously on 5-axis mills, holding critical spherical bearing seats to a ±0.05 mm diametric tolerance to prevent suspension slop under 5G cornering loads.

The stressed 590 mm longitudinal gearbox casing, cast from an advanced aluminium-lithium alloy, is subjected to post-cast 5-axis machining to hold the primary gear shaft bearing bores to a ±0.02 mm concentricity tolerance, ensuring the 350 kW MGU-K torque delivery does not induce catastrophic gear misalignment. The Carbon-Carbon brake discs are heavily diamond ground to ensure a ±0.1 mm thickness uniformity across the braking annulus, preventing high-speed brake judder. Every single safety-critical structural fastener is torque-verified manually and mechanically wire-locked to prevent vibrational backing-out.

13.4 Quality Control & Sign-Off

General vehicle assembly cannot commence without exhaustive Quality Control (QC) sign-offs. This includes a 100% ultrasonic NDT scan of all Class-A structural carbon elements and extensive CMM verification of the completed rolling chassis to ensure the global suspension points mathematically align with the static vehicle dynamics model.

The completed 4 MJ battery pack undergoes a full, aggressive charge-discharge cycle test protocol *ex-situ* to validate the active liquid cooling loop functionality and BMS cell balancing logic before it is physically bolted into the car. Finally, immediately prior to the installation of the protective outer bodywork, the entire High-Voltage (HV) tractive system is subjected to a final isolation test: the resistance between the active 648 V DC link and the grounded carbon chassis must definitively exceed >500 MΩ before the vehicle is legally signed off for track operation.

14 Conclusion

The impending 2026 Formula 1 regulations explicitly force a paradigm shift transcending incremental aerodynamic evolution. This paper has comprehensively drafted, derived, and mathematically validated a complete macroscopic vehicle architecture capable of simultaneously satisfying the aggressive mechanical, thermal, and electrical constraints of the new era.

14.1 Design Summary

Synthesizing the derived specifications establishes a cohesive vehicle platform. The architecture mandates a 3580 mm wheelbase and a strictly constrained 5125 mm overall length, bounded within the 800 kg minimum mass target. The global 270 mm Center of Gravity (CoG) is rigidly anchored by the 45/55 front-to-rear static weight distribution.

Every engineering parameter throughout this architecture is inextricably interconnected. The necessity to package the 180S2P, 648 V Nickel Manganese Cobalt (NMC) battery low in the chassis dictates the geometric CoG. The extreme 180 C peak transient discharge currents generate profound I^2R thermal loads, directly driving the multi-loop, liquid cold-plate thermal packaging. This immense cooling requirement dictates the internal capacity of the minimalist undercut sidepod philosophy, which ultimately defines the external aerodynamic boundary layer feeding the rear diffuser. The powertrain relies seamlessly on Field Oriented Control (FOC) governed by Space Vector Pulse Width Modulation (SVPWM) firing SiC inverters, completely reliant on an active Extended Kalman Filter for real-time State of Charge (SOC) estimation.

14.2 Simulation Validation

The mathematical validity of this cohesive architecture was decisively proven via the custom MATLAB point-mass simulation. Navigating the 5.793 km Autodromo Nazionale di Monza, the integrated aerodynamic, mechanical, and electrical parameters computed a blistering theoretical lap time of 1:16.79.

This establishes a profound ∼2.3-second improvement over the existing 1:19.119 circuit lap record. This dramatic acceleration is completely consistent with the aggressive 350 kW Motor Generator Unit-Kinetic (MGU-K) violently overpowering the previous 120 kW electrical output on corner exits. The simulation verified a maximum speed of 348.2 km/h, while strictly adhering to the hard 3.0 MJ operational energy budget. The dynamic SOC trace bottomed out at a perfectly healthy 25.3%, conclusively validating that the tiered deployment logic successfully prevents catastrophic battery degradation without artificially castrating terminal straight-line speed.

14.3 2026 as an Inflection Point

Historically, hybrid technology in Formula 1 served strictly as an ancillary support mechanism to the dominant internal combustion engine [20]. The 2026 regulations represent the definitive historical inflection point: the formula has transitioned into a genuinely hybrid-electric paradigm.

The MGU-K now contributes a staggering 46% of total peak vehicle power. Paired with the mandatory 100% sustainable drop-in fuel, the carbon loop is physically closed. Furthermore, the complete regulatory eradication of the MGU-H (Motor Generator Unit-Heat) forces the entire geometric burden of turbo-lag mitigation directly onto complex ICE tuning matrices and instantaneous MGU-K low-RPM torque filling. Consequently, the high-voltage electrical architecture—encompassing the SiC inverter topologies, battery cell chemistry selections, and deterministic motor control algorithms—has utterly usurped aerodynamics as the primary engineering battleground for grid supremacy.

14.4 Model Limitations

While the synthesized simulation fundamentally validates the architectural philosophy, academic rigor demands an explicit acknowledgment of localized modeling limitations. The quasi-steady-state point mass model inherently ignores the nonlinear realities of transient tire degradation and thermal blistering. It assumes constant mechanical grip, failing to account for suspension kinematic compliance under 5G cornering loads or the mass reduction as the 70 kg fuel load is burned throughout the lap.

Furthermore, the aerodynamic mapping utilized steady-state lift and drag coefficients (C_l and C_d) derived from literature references [21], bypassing original dynamic Computational Fluid Dynamics (CFD) mapping that would accurately capture severe downforce variations induced by transient yaw and pitch angles. Finally, the energy deployment logic assumes a constant thermodynamic efficiency chain, ignoring how electrical resistances fiercely spike alongside die junction temperatures.

14.5 Future Work

To graduate this theoretical model into a viable, race-engineering tool, future computational expansions must violently scale in three primary directions:

1. **Multi-Lap Simulation:** The point-mass optimization solver must be expanded iteratively across a full 77-lap race distance. This will mandate modeling transient fuel burn mass-loss, compounding tire degradation, and strategic, long-term SOC state-machine management across multi-stint race strategies. 2. **Multi-Track Validation:** The fundamental architecture must be validated across the extreme aerodynamic spectrum. Simulating the ultra-high downforce confines of Monaco versus the low-drag efficiency demands of Spa-Francorchamps will definitively quantify the localized aerodynamic setup deltas required between X-mode and Z-mode configurations. 3. **Higher Fidelity Dynamics:** The rudimentary point-mass model must be wholly superseded by an integrated 14 Degree-of-Freedom (DOF) multi-body vehicle dynamics solver. Coupling genuine RANS CFD aero-maps with a transient Pacejka Magic Formula tire model will allow the implementation of advanced Model Predictive Control (MPC) algorithms to autonomously govern the MGU-K energy deployment in real-time.

Ultimately, the 2026 Formula 1 regulations irrevocably declare that mechanical and aerodynamic engineering alone are no longer sufficient to secure a World Championship. Total mastery of complex power electronics, advanced battery electrochemistry, and deterministic digital motor control now sits as the absolute cornerstone of modern motorsport dominance.

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